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(54) **PESTICIDAL COMPOSITIONS AND
RELATED METHODS**

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(57) **ABSTRACT**

Pesticidal compositions are provided that include at least
one pesticidal active ingredient and at least one organic
solvent. Adjuvant compositions are provided for use with at
least one pesticidal active ingredient that include at least one
organic solvent. Methods of producing an adjuvant compo-
sition and pesticidal composition are also provided.

6 Claims, No Drawings

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PESTICIDAL COMPOSITIONS AND RELATED METHODS

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. Ser. No. 16/786,846, filed Feb. 10, 2020 which is a continuation-in-part of U.S. Ser. No. 16/185,578 filed Nov. 9, 2018 which claims priority to U.S. provisional patent application Ser. No. 62/583,626, filed Nov. 9, 2017 and U.S. provisional patent application Ser. No. 62/643,215, filed Mar. 15, 2018.

BACKGROUND OF INVENTION

The control of various pests such as, for example, fungal infestations and bacterial infections continues to be a major problem in pesticidal and horticultural fields. Particularly, genetic changes in pathogenic bacteria and fungi have generated resistance to the efficacy of many types of active ingredients. Fungicide and bactericide resistance have particularly become a major challenge rendering some pesticidal compositions containing a fungicide or bactericide less efficacious against certain fungi and bacteria. In an attempt to overcome these challenges, the end-user normally adds an adjuvant to the spray mixture to improve the performance of pesticides, however, such adjuvants may increase the cost and impact crop yield as well as the environment. Thus, there remains a need for low toxicity and crop safe adjuvant compositions that increase the efficacy of pesticidal active ingredients.

SUMMARY OF INVENTION

According to one aspect, a pesticidal composition is provided. The pesticidal composition includes at least one pesticidal active ingredient and at least one organic solvent such as propylene carbonate, N-methylpyrrolidone, benzyl acetate, benzyl butyrate, benzyl propionate, diethyl malonate, 3-methoxy-3-methyl-1-butanol, dimethyl benzyl carbinol acetate, phenyl ethyl alcohol, terpinyl acetate, benzyl benzoate, methyl salicylate, hexyl acetate, benzyl butyrate, diethyl malonate, ethyl acetate, phenoxy ethyl iso-butyrate, terpineol, geranyl acetate, linalyl acetate, ethyl 2-methyl butyrate, propylene glycol diacetate, dipropylene glycol monomethyl ether, and any combination thereof. According to one embodiment, the at least one organic solvent is benzyl acetate. According to one embodiment, the at least one organic solvent in the pesticidal compositions as provided herein is benzyl acetate, benzyl alcohol, phenyl ethyl alcohol, hexyl acetate, terpinyl acetate, terpineol, geranyl acetate, linalyl acetate, propylene glycol diacetate, benzyl propionate, dipropylene glycol monomethyl ether, propylene carbonate, 3-Methoxy-3-methyl-1-butanol, N-methylpyrrolidone, or any combination thereof. According to one embodiment, the pesticidal composition increases the efficacy of the at least one pesticidal active ingredient compared to the at least one pesticidal active ingredient alone. According to one embodiment, the pesticidal composition increases the efficacy of the at least one pesticidal active ingredient by at least about 5%, 10%, 15%, 20% 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more compared to the at least one pesticidal active ingredient alone. According to another embodiment, the pesticidal composition increases the efficacy of the at least one pesticidal active ingredient against a resistant or susceptible fungal strain compared to the at least

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one pesticidal active ingredient alone. According to another embodiment, the pesticidal composition increases the efficacy of the at least one pesticidal active ingredient against a resistant or susceptible fungal strain by at least about 5%, 10%, 15%, 20% 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more compared to the at least one pesticidal active ingredient alone.

According to one embodiment, the pesticidal composition further includes water, citric acid, potassium hydroxide, cocodimethylamine, polyoxyethylene sorbitan monolaurate, at least one emulsifier compound, at least one antifoaming compound, at least one antimicrobial agent, at least one buffer, at least one corrosion inhibitor, at least one dispersing agent, at least one dye, at least one fragrant, at least one freezing point depressant, at least one neutralizing agent, at least one odorant, at least one penetration aid, at least one sequestering agent, at least one spray drift control agent, at least one spreading agent, at least one stabilizer, at least one sticking agent, at least one viscosity-modifying additive, at least one preservative, at least one plasticizer, at least one hydrophobic barrier, at least one UV protectant, at least one water soluble solvent, or any combination thereof. According to one embodiment, the at least one pesticidal active ingredient is a fungicide, bactericide, herbicide, insecticide, pesticide, plant growth regulator, or biopesticide. According to one embodiment, the at least one pesticidal active ingredient is a fungicide. According to one embodiment, the fungicide is benzimidazole carbamates, dicarboximides, demethylation inhibitors, phenyl amides, morpholines, phosphothilates, dithiolanes, carboxamides (succinate dehydrogenase inhibitors), hydroxy-(2-amino)-pyrimidines, anilino-pyrimidines, N-phenyl carbamates, quinone outside inhibitors, phenylpyrroles, quinolines, aromatic hydrocarbons, heteroaromatics, melanin biosynthesis inhibitors, hydroxyanilides, thiocarbamates, allylamines, polyoxins, phenylureas, quinone inside inhibitors, benzamides, enopyranuronic acid antibiotic, hexopyranosyl antibiotic, glucopyranosyl antibiotic, cyanoacetamide-oximes, carbamates, dinitrophenyl crotonates, pyrimidine-hydrazones, 2,6-dinitro-anilines, organo tin compounds, carboxylic acids, heteroaromatics, phosphonates, phthalamic acids, benzotriazines, benzene-sulfonamides, pyridazinones, thiophene-carboxamides, pyrimidinamides, carboxylic acid amides, tetracycline antibiotic, thiocarbamate, benzo-thiadiazole, benzisothiazole, thiadiazole-carboxamide, thiazole-carboxamides, benzamidoxime, quinazolinone, benzophenone, acylpicolide, copper, sulfur, dithiocarbamates, phthalimides, chloronitriles, sulphamides, guanidines, triazines, quinones, mineral oils, organic oils, and potassium bicarbonate.

According to one embodiment, the pesticidal composition includes from about 1% w/w to about 70% w/w of the at least one pesticidal active ingredient, from about 39% w/w to about 93% w/w of the at least one organic solvent; and from about 2% w/w to about 80% w/w of water, citric acid, potassium hydroxide, cocodimethylamine, polyoxyethylene sorbitan monolaurate, at least one emulsifier compound, at least one antifoaming compound, at least one antimicrobial agent, at least one buffer, at least one corrosion inhibitor, at least one dispersing agent, at least one dye, at least one fragrant, at least one freezing point depressant, at least one neutralizing agent, at least one odorant, at least one penetration aid, at least one sequestering agent, at least one spray drift control agent, at least one spreading agent, at least one stabilizer, at least one sticking agent, at least one viscosity-modifying additive, at least one preservative, at least one plasticizer, at least one hydrophobic barrier, at least

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one UV protectant, at least one water soluble solvent, or any combination thereof, citric acid, potassium hydroxide, cocodimethylamine, polyoxyethylene sorbitan monolaurate, at least one emulsifier compound, at least one antifoaming compound, at least one antimicrobial agent, at least one buffer, at least one corrosion inhibitor, at least one dispersing agent, at least one dye, at least one fragrant, at least one freezing point depressant, at least one neutralizing agent, at least one odorant, at least one penetration aid, at least one sequestering agent, at least one spray drift control agent, at least one spreading agent, at least one stabilizer, at least one sticking agent, at least one viscosity-modifying additive, at least one preservative, at least one plasticizer, at least one hydrophobic barrier, at least one UV protectant, at least one water soluble solvent, or any combination thereof.

According to one aspect, an adjuvant composition is provided. The adjuvant composition includes at least one organic solvent such as propylene carbonate, N-methylpyrrolidone, benzyl acetate, benzyl butyrate, benzyl propionate, diethyl malonate, 3-methoxy-3-methyl-1-butanol, dimethyl benzyl carbinol acetate, phenyl ethyl alcohol, terpinyl acetate, benzyl benzoate, methyl sailcylate, hexyl acetate, benzyl butyrate, diethyl malonate, ethyl acetate, phenoxy ethyl iso-butyrate, terpineol, geranyl acetate, linalyl acetate, ethyl 2-methyl butyrate, propylene glycol diacetate, dipropylene glycol monomethyl ether, and any combination thereof. According to one embodiment, the at least one organic solvent in the adjuvant compositions as provided herein is benzyl acetate, benzyl alcohol, phenyl ethyl alcohol, hexyl acetate, terpinyl acetate, terpineol, geranyl acetate, linalyl acetate, propylene glycol diacetate, benzyl propionate, dipropylene glycol monomethyl ether, propylene carbonate, 3-Methoxy-3-methyl-1-butanol, N-methylpyrrolidone, or any combination thereof. The adjuvant composition, when combined with at least one pesticidal active ingredient, increases the efficacy of the at least one pesticidal active ingredient compared to the at least one pesticidal active ingredient alone. According to one embodiment, the adjuvant composition, when combined with at least one pesticidal active ingredient, increases the efficacy of the at least one pesticidal active ingredient by at least about 5%, 10%, 15%, 20% 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more compared to the at least one pesticidal active ingredient alone. According to another embodiment, the adjuvant composition, when combined with at least one pesticidal active ingredient, increases the efficacy of the at least one pesticidal active ingredient against a resistant or susceptible fungal strain compared to the at least one pesticidal active ingredient alone. According to another embodiment, the adjuvant composition, when combined with at least one pesticidal active ingredient, increases the efficacy of the at least one pesticidal active ingredient against a resistant or susceptible fungal strain by at least about 5%, 10%, 15%, 20% 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more compared to the at least one pesticidal active ingredient alone. According to one embodiment, the at least one organic solvent is benzyl acetate. According to one embodiment, the adjuvant composition further includes water, citric acid, potassium hydroxide, cocodimethylamine, polyoxyethylene sorbitan monolaurate, at least one emulsifier compound, at least one antifoaming compound, at least one antimicrobial agent, at least one buffer, at least one corrosion inhibitor, at least one dispersing agent, at least one dye, at least one fragrant, at least one freezing point depressant, at least one neutralizing agent, at least one odorant, at least one penetration aid, at

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least one sequestering agent, at least one spray drift control agent, at least one spreading agent, at least one stabilizer, at least one sticking agent, at least one viscosity-modifying additive, at least one preservative, at least one plasticizer, at least one hydrophobic barrier, at least one UV protectant, at least one water soluble solvent, or any combination thereof. According to one embodiment, adjuvant includes about 0.0% w/w to about 2.0% w/w water, about 0.0% w/w to about 2.0% w/w potassium hydroxide, about 0.0% w/w to about 2.0% w/w citric acid, about 0.0% w/w to about 2.0% w/w cocodimethylamine, about 20% w/w to about 99% w/w of at least one organic solvent, and about 10% w/w to about 70% w/w of at least one surfactant. According to one embodiment, the at least one organic solvent is benzyl acetate, propylene carbonate, N-methylpyrrolidone, or any combination thereof. According to one embodiment, the at least one surfactant is cocodimethylamine, polyoxyethylene sorbitan monolaurate. According to one embodiment, the at least one pesticidal active ingredient is a fungicide, bactericide, herbicide, insecticide, pesticide, plant growth regulator, or biopesticide. According to one embodiment, the at least one pesticidal active ingredient is a fungicide.

According to another aspect, a method of producing an adjuvant composition is provided. The method includes the step of introducing at least one organic solvent to a container. The at least one organic solvent includes propylene carbonate, N-methylpyrrolidone, benzyl acetate, benzyl butyrate, benzyl propionate, diethyl malonate, 3-methoxy-3-methyl-1-butanol, dimethyl benzyl carbinol acetate, phenyl ethyl alcohol, terpinyl acetate, benzyl benzoate, methyl sailcylate, hexyl acetate, benzyl butyrate, diethyl malonate, ethyl acetate, phenoxy ethyl iso-butyrate, terpineol, geranyl acetate, linalyl acetate, ethyl 2-methyl butyrate, propylene glycol diacetate, dipropylene glycol monomethyl ether, or any combination thereof. According to one embodiment, the at least one organic solvent is benzyl acetate, benzyl alcohol, phenyl ethyl alcohol, hexyl acetate, terpinyl acetate, terpineol, geranyl acetate, linalyl acetate, propylene glycol diacetate, benzyl propionate, dipropylene glycol monomethyl ether, propylene carbonate, 3-Methoxy-3-methyl-1-butanol, N-methylpyrrolidone, or any combination thereof. The method may further include the step of introducing at least one additional component to the container, wherein the at least one additional component is water, citric acid, potassium hydroxide, cocodimethylamine, polyoxyethylene sorbitan monolaurate, at least one emulsifier compound, at least one antifoaming compound, at least one antimicrobial agent, at least one buffer, at least one corrosion inhibitor, at least one dispersing agent, at least one dye, at least one fragrant, at least one freezing point depressant, at least one neutralizing agent, at least one odorant, at least one penetration aid, at least one sequestering agent, at least one spray drift control agent, at least one spreading agent, at least one stabilizer, at least one sticking agent, at least one viscosity-modifying additive, at least one preservative, at least one plasticizer, at least one hydrophobic barrier, at least one UV protectant, at least one water soluble solvent, or any combination thereof. The method may further include the step of attaching at least one set of instructions to the container. The at least one organic solvent is introduced to the container in an amount such that the entire container contents may be added directly to a spray tank comprising at least one pesticidal active ingredient for form a tank mix pesticidal composition. According to one embodiment, the at least one organic solvent is benzyl acetate. According to one embodiment, the adjuvant composition increases the efficacy of the at least one pesticidal active ingredient by at least about 25%

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compared to the at least one pesticidal active ingredient alone. According to one embodiment, the at least one pesticidal active ingredient is a fungicide, bactericide, herbicide, insecticide, pesticide, plant growth regulator, or biopesticide. According to another embodiment, the adjuvant composition, when combined with at least one pesticidal active ingredient, increases the efficacy of the at least one pesticidal active ingredient against a resistant or susceptible fungal strain compared to the at least one pesticidal active ingredient alone.

According to one aspect, a tank mix pesticidal composition is provided. The tank mix pesticidal composition includes at least one pesticidal active ingredient, at least one organic solvent, and water, spray oil, or a combination of water and spray oil. The at least one organic solvent is propylene carbonate, N-methylpyrrolidone, benzyl acetate, benzyl butyrate, benzyl propionate, diethyl malonate, 3-methoxy-3-methyl-1-butanol, dimethyl benzyl carbinol acetate, phenyl ethyl alcohol, terpinyl acetate, benzyl benzoate, methyl sailcylate, hexyl acetate, benzyl butyrate, diethyl malonate, ethyl acetate, phenoxy ethyl iso-butyrate, terpineol, geranyl acetate, linalyl acetate, ethyl 2-methyl butyrate, propylene glycol diacetate, dipropylene glycol monomethyl ether, or any combination thereof. According to one embodiment, the at least one organic solvent in the tank mix as provided herein is benzyl acetate, benzyl alcohol, phenyl ethyl alcohol, hexyl acetate, terpinyl acetate, terpineol, geranyl acetate, linalyl acetate, propylene glycol diacetate, benzyl propionate, dipropylene glycol monomethyl ether, propylene carbonate, 3-Methoxy-3-methyl-1-butanol, N-methylpyrrolidone, or any combination thereof. According to one embodiment, the tank mix includes from about 0.01% w/w to about 5% w/w of at least one pesticidal active ingredient; from about 0.2% w/w to about 2 w/w % of at least one organic solvent; and from about 80% w/w to about 99% of water, spray oil, or a combination of water and spray oil. According to one embodiment, the at least one active ingredient is a fungicide, bactericide, herbicide, insecticide, pesticide, or biopesticide. According to one embodiment, the organic solvent is benzyl acetate. The tank mix pesticidal composition increases the efficacy of the at least one pesticidal active ingredient compared to the at least one pesticidal active ingredient alone. According to one embodiment, the tank mix pesticidal composition increases the efficacy of the at least one pesticidal active ingredient by at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more compared to the at least one pesticidal active ingredient alone. According to another embodiment, the tank mix pesticidal composition increases the efficacy of the at least one pesticidal active ingredient against a resistant or susceptible fungal strain compared to the at least one pesticidal active ingredient alone. According to another embodiment, the tank mix pesticidal composition increases the efficacy of the at least one pesticidal active ingredient against a resistant or susceptible fungal strain by at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more compared to the at least one pesticidal active ingredient alone.

According to one aspect, an adjuvant composition for increasing the efficacy of a fungicidal active ingredient is provided. The adjuvant composition increases the efficacy of any fungicidal or bactericide active ingredient as provided in one or more FRAC Groups. According to one embodiment, the adjuvant composition increases efficacy against fungicidal resistant strains of fungi. According to one embodiment, the adjuvant composition increases efficacy against

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fungicidal susceptible strains of fungi. According to one embodiment, the adjuvant composition includes at least one organic solvent is propylene carbonate, N-methylpyrrolidone, benzyl acetate, benzyl butyrate, benzyl propionate, diethyl malonate, 3-methoxy-3-methyl-1-butanol, dimethyl benzyl carbinol acetate, phenyl ethyl alcohol, terpinyl acetate, benzyl benzoate, methyl sailcylate, hexyl acetate, benzyl butyrate, diethyl malonate, ethyl acetate, phenoxy ethyl iso-butyrate, terpineol, geranyl acetate, linalyl acetate, ethyl 2-methyl butyrate, propylene glycol diacetate, dipropylene glycol monomethyl ether, or any combination thereof. According to one embodiment, the adjuvant composition includes at least one benzyl acetate, benzyl alcohol, phenyl ethyl alcohol, hexyl acetate, terpinyl acetate, terpineol, geranyl acetate, linalyl acetate, propylene glycol diacetate, benzyl propionate, dipropylene glycol monomethyl ether, propylene carbonate, 3-Methoxy-3-methyl-1-butanol, N-methylpyrrolidone, or any combination thereof. The adjuvant composition, when combined with at least one fungicidal or bactericide active ingredient, increases the efficacy of the at least one fungicidal or bactericide active ingredient against fungicidal or bactericide resistant or susceptible strains compared to the at least one fungicidal or bactericide active ingredient alone. According to one embodiment, the fungicidal or bactericide active ingredient is selected from the group consisting of methyl benzimidazole carbamates, succinate dehydrogenase inhibitors, quinone outside inhibitors, demethylation inhibitors, and anilino-pyrimidines. According to one embodiment, the fungicidal or bactericide active ingredient falls within one or more of the FRAC groups that include Groups 1-15, 16.1, 16.2, 17-42, P, U, M, and NC. According to one embodiment, the fungicidal or bactericide active ingredient includes methyl benzimidazole carbamates, dicarboximides, demethylation inhibitors, phenyl amides, morpholines, phosphothilates, dithiolanes, carboxamides (succinate dehydrogenase inhibitors), hydroxy-(2-amino-)pyrimidines, anilino-pyrimidines, N-phenyl carbamates, quinone outside inhibitors, phenylpyrroles, quinolines, aromatic hydrocarbons, heteroaromatics, melanin biosynthesis inhibitors, hydroxyanilides, thiocarbamates, allylamines, polyoxins, phenylureas, quinone inside inhibitors, benzamides, enopyranuronic acid antibiotic, hexopyranosyl antibiotic, glucopyranosyl antibiotic, cyanoacetamide-oximes, carbamates, dinitrophenyl crotonates, pyrimidine-hydrazones, 2,6-dinitro-anilines, organo tin compounds, carboxylic acids, heteroaromatics, phosphonates, phthalamic acids, benzotriazines, benzene-sulfonamides, pyridazinones, thiophene-carboxamides, pyrimidinamides, carboxylic acid amides, tetracycline antibiotic, thiocarbamate, benzo-thiadiazole, benzisothiazole, thiadiazole-carboxamide, thiazole-carboxamides, benzamidoxime, quinazolinone, benzophenone, acylpicolide, copper, sulfur, dithiocarbamates, phthalimides, chloronitriles, sulphamides, guanidines, triazines, quinones, mineral oils, organic oils, and potassium bicarbonate. According to one embodiment, the adjuvant includes about 0.0% w/w to about 2.0% w/w water; about 0.0% w/w to about 2.0% w/w potassium hydroxide; about 0.0% w/w to about 2.0% w/w citric acid; about 0.0% w/w to about 2.0% w/w cocodimethylamine; about 20% w/w to about 99% w/w of at least one organic solvent; and about 10% w/w to about 70% w/w of at least one surfactant.

According to another embodiment, the adjuvant composition, when combined with at least one fungicidal active ingredient, increases the efficacy of the at least one fungicidal active ingredient against a resistant or susceptible fungal strain compared to the at least one fungicidal active

ingredient alone. According to another embodiment, the adjuvant composition, when combined with at least one fungicidal active ingredient, increases the efficacy of the at least one fungicidal active ingredient against a resistant or susceptible fungal strain by at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% compared to the at least one fungicidal active ingredient alone.

According to another aspect, an adjuvant composition for increasing the efficacy of a bactericide active ingredient is provided. The adjuvant composition includes at least one organic solvent selected from the group consisting of benzyl acetate, benzyl alcohol, phenyl ethyl alcohol, hexyl acetate, terpinyl acetate, terpineol, geranyl acetate, linalyl acetate, propylene glycol diacetate, benzyl propionate, dipropylene glycol monomethyl ether, propylene carbonate, 3-Methoxy-3-methyl-1-butanol, N-methylpyrrolidone, diethyl malonate, or any combination thereof. The adjuvant composition, when combined with at least one bactericide active ingredient, increases the efficacy of the at least one bactericide active ingredient compared to the at least one bactericide active ingredient alone. According to another embodiment, the adjuvant composition, when combined with at least one bactericide active ingredient, increases the efficacy of the at least one bactericide active ingredient against a resistant or susceptible bacterial strain by at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% compared to the at least one bactericide active ingredient alone. According to one embodiment, the at least one bactericide active ingredient includes phthalimide and dithiocarbamate and the at least one organic solvent is diethyl malonate.

DETAILED DESCRIPTION

The present disclosure will now be described more fully hereinafter with reference to exemplary embodiments thereof. These exemplary embodiments are described so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. Indeed, the present disclosure may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. As used in the specification, and in the appended claims, the singular forms "a", "an", "the", include plural referents unless the context clearly dictates otherwise.

As used herein, the term "adjuvant" refers to one or more organic solvents that improve the desired performance of a pesticidal active ingredient.

As used herein, the term "pesticide" or "pesticidal" refers to crop use, turf use, ornamental use, homeowner use or similar use.

As used herein, the term "FRAC" refers to the fungicide resistance action committee.

As used herein, the term "FRAC Group" refers to groups of a list of groups of fungicides and bactericides according to their mode of action and resistance risk. The FRAC groups are widely accepted in literature and incorporated herein by reference. The FRAC groups include Groups 1-15, 16.1, 16.2, 17-42, P, U, M, and NC. The FRAC groups include methyl benzimidazole carbamates, dicarboximides, demethylation inhibitors, phenyl amides, morpholines, phosphothilates, dithiolanes, carboxamides (succinate dehydrogenase inhibitors), hydroxy-(2-amino)-pyrimidines, anilino-pyrimidines, N-phenyl carbamates, quinone out-

side inhibitors, phenylpyrroles, quinolines, aromatic hydrocarbons, heteroaromatics, melanin biosynthesis inhibitors, hydroxylanilides, thiocarbamates, allylamines, polyoxins, phenylureas, quinone inside inhibitors, benzamides, enopyranuronic acid antibiotic, hexopyranosyl antibiotic, glucopyranosyl antibiotic, cyanoacetamide-oximes, carbamates, dinitrophenyl crotonates, pyrimidine-hydrazones, 2,6-dinitro-anilines, organo tin compounds, carboxylic acids, heteroaromatics, phosphonates, phthalamic acids, benzotriazines, benzene-sulfonamides, pyridazinones, thiophene-carboxamides, pyrimidinamides, carboxylic acid amides, tetracycline antibiotic, thiocarbamate, benzo-thiadiazole, benzisothiazole, thiadiazole-carboxamide, thiazole-carboxamides, benzamidoxime, quinazolinone, benzophenone, acylpicolide, copper, sulfur, dithiocarbamates, phthalimides, chloronitriles, sulphamides, guanidines, triazines, quinones, mineral oils, organic oils, and potassium bicarbonate.

Pesticidal compositions for improving the desired performance of a pesticidal active ingredient are provided. The pesticidal compositions described herein include at least one organic solvent, at least one active ingredient and, optionally, one or more additional components as provided herein. According to one embodiment, the pesticidal compositions as provided herein are added to a spray tank such that a stable tank mix pesticidal composition (i.e., spray tank mixture) is formed when the pesticidal composition and any other additional optional ingredients, such as water or spray oil, are combined in the spray tank.

According to one embodiment, the pesticidal compositions as provided herein are formulated as an emulsifiable concentrate formulation, aqueous suspension concentrate (suspension concentrate), emulsion in water formulation, dry flowable, water dispersible granule, wettable powder, soluble powder, oil dispersion, or concentrated aqueous solution.

An organic solvent as provided herein is preferably not phytotoxic, approved for crop use by the Environmental Protection Agency (EPA) and safe for end user mixing. Suitable organic solvents include, but are not limited to, organic compounds within the chemical classes of carbonates, ester compounds, amide compounds, ketone compounds, and alcohol compounds. According to a particular embodiment, the organic solvent is propylene carbonate, N-methylpyrrolidone, benzyl acetate, benzyl butyrate, benzyl propionate, diethyl malonate, 3-methoxy-3-methyl-1-butanol, dimethyl benzyl carbinol acetate, phenyl ethyl alcohol, terpinyl acetate, benzyl benzoate, methyl sailcylate, hexyl acetate, benzyl butyrate, diethyl malonate, ethyl acetate, phenoxy ethyl iso-butyrate, terpineol, geranyl acetate, linalyl acetate, ethyl 2-methyl butyrate, propylene glycol diacetate, dipropylene glycol monomethyl ether, or any combination thereof.

According to a particular embodiment, the at least one organic solvent provided herein is benzyl acetate. The benzyl acetate as provided herein may be commercially available from various sources such as, for example, Sigma Aldrich. The benzyl acetate as provided herein may also be derived from a natural source such as, for example, essential oils. Suitable essential oil sources include, but are not limited to, jasmine oil, ylang ylang oil, neroli oil, or any combination thereof.

According to a particular embodiment, the organic solvent is propylene carbonate. According to another particular embodiment, the organic solvent is N-methylpyrrolidone. According to one embodiment, the organic solvent is benzyl acetate, benzyl alcohol, phenyl ethyl alcohol, hexyl acetate, terpinyl acetate, terpineol, geranyl acetate, linalyl acetate,

propylene glycol diacetate, benzyl propionate, dipropylene glycol monomethyl ether, propylene carbonate, 3-Methoxy-3-methyl-1-butanol, N-methylpyrrolidone, or any combination thereof.

Other suitable organic solvents include, but are not limited to, acetate esters such as, for example, n-butyl acetate, iso-butyl acetate, n-pentyl acetate, iso-pentyl acetate, n-hexyl acetate, iso-hexyl acetate, cyclohexyl acetate, phenyl acetate, n-heptyl acetate, iso-heptyl acetate, cyclohexylmethyl acetate, benzyl alcohol, benzyl acetate, benzyl salicylate, toluene, xylene, petroleum ether, vegetable oils, acetone, methyl ethyl ketone, cyclohexanone, acid anhydrides, acetonitrile, acetophenone, amyl acetate, 2-butanone, butylenes carbonate, 1, 2-butylene carbonate, diethylene glycol butyl ether, diethylene glycol butyl ether acetate, dibasic ester mixture, dimethyl sulfoxide, dipropylene glycol methyl ether, ethylene glycol butyl ether, ethylene carbonate, 3-ethoxy ethyl propionate, ethylene glycol diacetate, furfuryl alcohol, gamma-butyrolactone, methyl ethyl ketone, methyl isoamyl ketone, n-amyl acetate, n-methyl-2-pyrrolidone, propylene glycol butyl ether, propylene carbonate, propylene, glycol methyl ether, propylene glycol methyl ether acetate, triethylene glycol, glycol diether, chlorobenzene, cyclohexane, cyclohexanol, alkyl esters of acetic acid, diacetone alcohol, 1,2-dichloropropane, diethanolamine, p-diethylbenzene, diethylene glycol, diethylene glycol abietate, diethylene glycol butyl ether, diethylene glycol ethyl ether, diethylene glycol methyl ether, N,N-dimethylformamide, dimethyl sulfoxide, 1,4-dioxane, dipropylene glycol, dipropylene glycol methyl ether, dipropylene glycol dibenzoate, diproxitol, alkylpyrrolidone, 2-ethyl hexanol, ethylene carbonate, 1,1,1-trichloroethane, 2-heptanone, alpha-pinene, d-limonene, ethyl lactate, ethylene glycol, ethylene glycol butyl ether, ethylene glycol methyl ether, gamma-butyrolactone, glycerol, glycerol acetate, glycerol diacetate, glycerol triacetate, hexadecane, hexylene glycol, isoamyl acetate, isobornyl acetate, isooctane, isophorone, isopropylbenzene, isopropyl myristate, lactic acid, laurylamine, mesityl oxide, methoxypropanol, methyl isoamyl ketone, methyl isobutyl ketone, methyl laurate, methyl octanoate, methyl oleate, methylene chloride, m-xylene, n-hexane, n-octylamine, octadecanoic acid, octylamine acetate, oleic acid, oleylamine, o-xylene, phenol, polyethylene glycol (PEG 400), propionic acid, propyl lactate, propylene carbonate, propylene glycol, propylene glycol methyl ether, p-xylene, toluene, triethyl phosphate, triethylene glycol, xylenesulfonic acid, paraffin, mineral oil, trichloroethylene, perchloroethylene, ethyl acetate, amyl acetate, butyl acetate, propylene glycol methyl ether, diethylene glycol methyl ether, methanol, ethanol, isopropanol, amyl alcohol, tetrahydrofurfuryl alcohol, hexanol, octanol, ethylene glycol, propylene glycol, glycerol, N-methyl-2-pyrrolidone, glycerol, pentaerythritol, dimethyl sulfoxide, dimethyl formamide, glymes, acetone, Atlox™ 4915, Atlox™ PN-100, Zephrym™ PD3315, any of the Tween™ L Series compounds, and combinations thereof.

According to one embodiment, the pesticidal compositions as provided herein include typically at least from about 39% by weight (w/w) to about 93% by weight (w/w) of at least one organic solvent as provided herein. According to one embodiment, the pesticidal composition includes typically at least about 45% by weight (w/w) of at least one organic solvent as provided herein. According to one embodiment, the pesticidal composition includes typically at least about 55% by weight (w/w) of at least one organic solvent as provided herein. According to one embodiment, the pesticidal composition includes typically at least about

65% by weight (w/w) of at least one organic solvent as provided herein. According to one embodiment, the pesticidal composition includes typically at least about 75% by weight (w/w) of at least one organic solvent as provided herein. According to one embodiment, the pesticidal composition includes typically at least about 85% by weight (w/w) of at least one organic solvent as provided herein. According to one embodiment, the pesticidal composition includes typically at least about 90% by weight (w/w) of at least one organic solvent as provided herein. According to one embodiment, the pesticidal composition includes typically at least about 93% by weight (w/w) of at least one organic solvent as provided herein.

According to one embodiment, the pesticidal compositions as provided herein include typically from about 5% by weight (w/w) to about 70% by weight (w/w) of the at least one pesticidal active ingredient as provided herein. According to one embodiment, the pesticidal composition includes typically at least about 10% by weight (w/w) of at least one pesticidal active ingredient as provided herein. According to one embodiment, the pesticidal composition includes typically at least about 20% by weight (w/w) of at least one pesticidal active ingredient as provided herein. According to one embodiment, the pesticidal composition includes typically at least about 30% by weight (w/w) of at least one pesticidal active ingredient as provided herein. According to one embodiment, the pesticidal composition includes typically at least about 40% by weight (w/w) of at least one pesticidal active ingredient as provided herein. According to one embodiment, the pesticidal composition includes typically at least about 50% by weight (w/w) of at least one pesticidal active ingredient as provided herein. According to one embodiment, the pesticidal composition includes typically at least about 60% by weight (w/w) of at least one pesticidal active ingredient as provided herein.

According to one embodiment, the pesticidal compositions as provided herein include typically from about 2% by weight (w/w) to about 80% by weight (w/w) of water, citric acid, potassium hydroxide, cocodimethylamine, polyoxyethylene sorbitan monolaurate, at least one emulsifier compound, at least one antifoaming compound, at least one antimicrobial agent, at least one buffer, at least one corrosion inhibitor, at least one dispersing agent, at least one dye, at least one fragrant, at least one freezing point depressant, at least one neutralizing agent, at least one odorant, at least one penetration aid, at least one sequestering agent, at least one spray drift control agent, at least one spreading agent, at least one stabilizer, at least one sticking agent, at least one viscosity-modifying additive, at least one preservative, at least one plasticizer, at least one hydrophobic barrier, at least one UV protectant, at least one water soluble solvent, or any combination thereof.

As provided herein, a pesticidal active ingredient is a fungicide, bactericide, herbicide, insecticide, pesticide, biopesticide, plant growth regulator or any combination thereof. The at least one pesticidal active ingredient may be any active ingredient approved by the Environmental Protection Agency and registered under the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA).

According to one embodiment, the at least one pesticidal active ingredient is a fungicide or bactericide. According to one embodiment, the fungicide or bactericide is a FRAC Group 1-15, 16.1, 16.2, 17-42, P, U, M, or NC fungicide. The FRAC groups include methyl benzimidazole carbamates, dicarboximides, demethylation inhibitors, phenyl amides, morpholines, phosphothilates, dithiolanes, carboxamides (succinate dehydrogenase inhibitors), hydroxy-(2-

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amino-)pyrimidines, anilino-pyrimindines, N-phenyl carbamates, quinone outside inhibitors, phenylpyrroles, quinolines, aromatic hydrocarbons, heteroaromatics, melanin biosynthesis inhibitors, hydroxyanilides, thiocarbamates, allylamines, polyoxins, phenylureas, quinone inside inhibitors, benzamides, enopyranuronic acid antibiotic, hexopyranosyl antibiotic, glucopyranosyl antibiotic, cyanoacetamide-oximes, carbamates, dinitrophenyl crotonates, pyrimidine-hydrazones, 2,6-dinitro-anilines, organo tin compounds, carboxylic acids, heteroaromatics, phosphonates, phthalamic acids, benzotriazines, benzene-sulfonamides, pyridazinones, thiophene-carboxamides, pyrimidinamides, carboxylic acid amides, tetracycline antibiotic, thiocarbamate, benzo-thiadiazole, benzisothiazole, thiadiazole-carboxamide, thiazole-carboxamides, benzamidoxime, quinazolinone, benzophenone, acylpicolide, copper, sulfur, dithiocarbamates, phthalimides, chloronitriles, sulphamides, guanidines, triazines, quinones, mineral oils, organic oils, and potassium bicarbonate.

According to one embodiment, the fungicide is within the family of strobilurin fungicides. The strobilurin fungicides as provided herein include all fungicides within FRAC Group 11. According to a particular embodiment, the at least one pesticidal active ingredient is a fungicide such as, for example, azoxystrobin. According to another embodiment, the at least one pesticidal active ingredient is a fungicide such as, for example, iprodione, oxytetracycline, biflunoxazole, coumethoxystrobin, coumoxystrobin, dimoxystrobin, enestroburin, enoxastrobin, fenaminstrobin, fenoxystrobin, flufenoxystrobin, fluoxastrobin, jiaxiangjunzhi, kresoxim-methyl, mandestrobin, metominostrobin, orysastrobin, picoxystrobin, pyraclostrobin, pyrametostrobin, pyraoxystrobin, triclopyricarb, trifloxystrobin, methyl 2-[2-(2,5-dimethylphenyloxymethyl)phenyl]-3-methoxyacrylate, pyribencarb, triclopyricarb/chlorodincarb, famoxadon, fenamidon, cyazofamid, amisulbrom, benodanil, bixafen, boscalid, carboxin, fenfuram, fluopyram, flutolanil, fluxapyroxad, furametpyr, isopyrazam, mepronil, oxycarboxin, penflufen, penthiopyrad, sedaxane, teclotalam, thifluzamide, N-(4'-trifluoromethylthio-biphenyl-2-yl)-3-difluoromethyl-1-methyl-1H-pyrazole-4-carboxamide, N-(2-(1,3,3-trimethylbutyl)phenyl)-1,3-dimethyl-5-fluoro-1H-pyrazole-4-carboxamide, N-[9-(dichloromethylene)-1,2,3,4-tetrahydro-1,4-methanonaphthalen-5-yl]-3-(difluoromethyl)-1-methyl-1H-pyrazole-4-carboxamide, diflumetorim, binapacryl, dinobuton, dinocap, meptyl-dinocap, fluazinam, ferimzone, ametocradin, silthiofam, azaconazole, bitertanol, bromconazole, cyproconazole, difenoconazole, diniconazole, diniconazole-M, epoxiconazole, fenbuconazole, fluquinconazole, flusilazole, flutriafol, hexaconazole, imibenconazole, ipconazole, metconazole, myclobutanil, oxpoconazole, paclobutrazole, penconazole, propiconazole, prothioconazole, simeconazole, tebuconazole, tetraconazole, triadimefon, triadimenol, triticonazole, uniconazole, imazalil, pefurazate, prochloraz, triflumizole, pyrimidines, fenarimol, nuarimol, pyrifenoxy, triforine, aldimorph, dodemorph, dodemorph acetate, fenpropimorph, tridemorph, fenpropidin, piperalin, spiroxamine, fenhexamid, benalaxyl, benalaxyl-M, kiralaxyl, metalaxyl, metalaxyl-M (mefenoxam), ofurace, oxadixyl, hymexazole, oethilone, oxolinic acid, bupirimate, benomyl, carbendazim, fuberidazole, thiabendazole, thiophanate-methyl, 5-chloro-7-(4-methyl-piperidin-1-yl)-6-(2,4,6-trifluorophenyl)-[1,2,4]triazolo[1,5-a]pyrimidine, diethofencarb, ethaboxam, pencycuron, fluopicolid, zoxamid, metrafenon, pyriofenon, cyprodinil, mepanipyrim, pyrimethanil, fluoroimide, iprodione, procymidone, vinclozolin, fenpiclonil, fludioxonil, quinoxifen,

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edifenphos, iprobenfos, pyrazophos, isoprothiolane, dicloran, quintozone, tecnazene, tolclofos-methyl, biphenyl, chloroneb, etridiazole, dimethomorph, flumorph, mandipropamid, pyrimorph, benthiavalicarb, iprovalicarb, valifenalate and 4-fluorophenyl N-(1-(1-(4-cyanophenyl)ethanesulfonyl)but-2-yl)carbamate, propamocarb, propamocarb hydrochloride, ferbam, mancozeb, maneb, metiram, propineb, thiram, zineb, ziram, anilazine, chlorothalonil, captafol, captan, folpet, dichlofluanid, dichlorophen, flusulfamide, hexachlorobenzene, pentachlorophenol, phthalid, tolylfluanid, N-(4-chloro-2-nitrophenyl)-N-ethyl-4-methylbenzenesulfonamide, guanidine, dithianon, validamycin, polyoxin B, pyroquilon, tricyclazole, carpropamid, dicyclomet, fenoxanil, and mixtures thereof.

According to one embodiment, the at least one pesticidal active ingredient is a herbicide. Suitable herbicides include those from all of the major families of herbicide chemistry including but not limited to sulfonylurea, triazine, uracil, imidazoline, pyridine, and urea. Suitable herbicides include but are not limited to, 2, 4-D, clyopyralid, dicamba, fosamine, glyphosate, imazapyr, hexazinone, imazapyr, metsulfuron methyl, pchloram, sulfometuron methyl and other sulfonylurea compounds, triclopyr, ethofumesate, bifenazate, metribuzin, fenmedipham, desmedipham, thidiazuron, diuron, quinclorac, clodinafop, fenoxaprop, tralkoxydim, prosulfocarb, triasulfuron, prosulfuron, amidosulfuron, iodosulfuron, chlorsulfuron, flupyralsulfuron, mesosulfuron, metsulfuron, sulfosulfuron, thifensulfuron, tribenuron, tritosulfuron, florasulam, metosulam, flumetsulam, pyroxulam, dichlorprop-p, MCPA, mecoprop, mecoprop-p, MCPB, clopyralid, bromoxynil, bromoxynil-octanoate, ioxynil, ioxynil-octanoate, fluroxypyr, trifluralin, diflufenican, picolinafen, pendimethalin and triallate, tralkoxydim, triasulfuron, diflufenican, florasulam, pyroxulam, pyroxulam, bipyridyliums, diphenyl ethers (nitrophenyl ethers), triazines, uracils, phenylureas, nitriles, Stam (3',4'-dichloropropionanilide, DCPA) and Alachlor (2-chloro-2',6'-diethyl-N-(methoxymethyl)-acetanilide), DCMU (3-(3,4-dichlorophenyl)-1,1-dimethylurea) and Rinuron (3-(3,4-dichlorophenyl)-1-methoxy-1-methylurea), thifensulfuronmethyl (methyl-3-(4-methoxy-6-methyl-1,3,5-triazin-2-ylcarbamoylsulfamoyl)-2-tanoate), Flazasulfuron (1-(4,6-dimethoxy pyrimidin-2-yl)-3-(3-trifluoromethyl-2-pyridylsulfonyl)urea), Paraquat dichloride (1,1'-dimethyl-4,4'-bipyridinium dichloride), Diquat dibromide (6,7-dihydrodipyridyl[1,2-a:2',1'-c]-pyrazinedium dibromide), Bromacil (5-bromo-3-sec-butyl-6-methyluracil), Gesatop (2-chloro-4,6-bis(ethylamino)-1,3,5-triazine), Simetryn (2,4-bis(ethylamino)-6-methylthio-1,3,5-triazine), DBN (2,6-dichlorobenzonitrile-), Trifluralin (alpha,alpha, alpha-trifluoro-2,6-dinitro-N,N-dipropyl-p-toluidine), Thiobencarb (Saturn) (S-p-chlorobenzyl diethylthiocarbamate), MCC (methyl-3,4-dichlorocarbonylate, NIP (2,4-dichlorophenyl-p-nitro-phenyl ether), PCP (sodium pentachlorophenoxide), MDBA (3,6-dichloro-2-methoxybenzoic acid dimethylamine salt), 2,4-D sodium salt (sodium 2,4-dichlorophenoxyacetate), 2,4 D Esters, Mapica ([4-chloro-o-toluyloxy]aceto-o-chloroanilide), Bialaphos (sodium salt of L-2-amino-4-[(hydroxy(methyl)phosphino)butyl]-alanyl-N-alanine), Glufosinate (ammonium DL-homoalanin-4-yl(methyl)phosphinate), TCA sodium salt (sodium trichloronate), and combinations thereof.

According to one embodiment, the at least one pesticidal active ingredient is an insecticide. Suitable insecticides include, but are not limited to, organochlorines, organophosphates, azinphos-methyl, azinphos-ethyl, chlorpyrifos, chlorpyrifos-methyl, chlorfenvinphos, diazinon, dimethylvin-

phos, dioxabenzofos, disulfoton, ethion, EPN, fenitrothion, fenthion, heptenophos, isoxathion, malathion, methidathion, methyl-parathion, paraoxon, parathion, phenthoate, phosalone, phosmet, phorate, phoxim, pirimiphos-methyl, profenofos, prothiofos, primiphos-ethyl, pyraclofos, pyridaphenthion, sulprofos, triazophos, trichlorfon, tetrachlorvinphos, vamidothion; carbamates, alanycarb, benfuracarb, bendiocarb, carbaryl, carbofuran, carbosulfan, fenoxycarb, furathiocarb, indoxacarb, methiocarb, pirimicarb, propoxur, thiodicarb, triazamate, organosulfurs, carbamates, formamidines, dinitrophenols, organotin, pyrethroids, acrinathrin, allethrin, bioallethrin, barthrin, bioethanomethrin, cyclothrin, bifenthrin, cyfluthrin, beta-cyfluthrin, cycloprothrin, cypermethrin, cyphenothrin, deltamethrin, dimethrin, empenethrin, esfenvalerate, etofenprox, flufenprox, halfenprox, profenbute, fenpirithrin, fenfluthrin, fenpropathrin, fenvalerate, flucythrinate, furethrin, imiprothrin, metofluthrin, cyhalothrin, lambda-cyhalothrin, gamma-cyhalothrin, permethrin, biopermethrin, transpermethrin, phenothrin, prallethrin, profluthrin, pyresmethrin, resmethrin, bioresmethrin, cismethrin, silafluofen, fluralinate, tau-fluvalinate, tefluthrin, terallethrin, tetramethrin, tralomethrin, alpha-cypermethrin, beta-cypermethrin, theta-cypermethrin, zeta-cypermethrin, permethrin, nicotinoids, spinosyns, pyrazoles, pyridazinones, quinazolines, abamectin, acequinocyl, acetamiprid, amitraz, azadirachtin, bensultap, bifenazate, cartap, chlorfenvap, chlordimeform, diafenthiuron, dinetofuran, diofenolan, emamectin, endosulfan, ethiprole, fenazaquin, fipronil, formetanate, formetanate hydrochloride, gamma-HCH, hydramethylnon, imidacloprid, indoxacarb, isoprocarb, metolcarb, pyridaben, pymetrozine, spinosad, tebufenpyrad, thiamethoxam, thiocyclam, pyridalyl, flonicamid, flucyprym, milbemectin, spiromesifen, flupyrzofos, tolfenpyrad, flubendiamide, bistrifluron, benclothiaz, pyrafluprole, pyriprole, amidoflumet, flufenimer, cyflumetofen, acequinocyl, lepimectin, profluthrin, dimethrin, xylylcarb, bromopropylate, spiroticlofen, clofentezine, fenpyroximate, hexythiazox, Fenvalerate (alpha-cyano-3-phenoxybenzyl-2-(4-chlorophenyl)-3-methylbutanoate), Baythroid (cyano-4-fluoro-3-phenoxybenzyl-3-(2,2-dichlorovinyl)-2,2-dimethylcyclopropanecarboxylate), DDVP (2,2-dichlorovinyl dimethyl phosphate), Sumithion (MEP) (dimethyl 4-nitro-m-tolyl phosphorothioate), Malathion (S-1,2-bis(ethoxycarbonyl)ethyl dimethyl phosphorodithioate), Dimethoate (dimethyl S-(N-methylcarbamoylmethyl) phosphorodithioate), Elsan (S-[alpha-(ethoxycarbonyl)benzyl]dimethyl phosphorodithioate), Baycid (dimethyl 4-methylthio-m-tolyl phosphorothioate), Bassa (O-sec-butylphenyl methylcarbamate), MTMC (m-tolylmethylcarbamate), Meopal (3,4-dimethylphenyl-N-methylcarbamate), NAC (1-naphthyl methylcarbamate), Methomyl (S-methyl-N-(methylcarbamoyloxy)thioacetimidate), Cartap (SS'-2-dimethylamino trimethylene bis-(thiocarbamate), cinnamon leaf oil, oregano oil, polyketide, pyrethrum, ryanodine, spinosad, spinosyn A, spinosyn D, thymol, *Bacillus sphaericus*, *Bacillus thuringiensis*, *Bacillus thuringiensis aizawi*, *Bacillus thuringiensis israelensis*, *Bacillus thuringiensis kurstaki*, *Bacillus thuringiensis tenebrionis*, nuclear polyhedrosis virus, granulovirus, *Lecanicillium lecanii*, diatomaceous earth, borate, borax, boric acid, and mixtures thereof.

According to one embodiment, the at least one pesticidal active ingredient is a bactericide. Suitable bactericides include any pesticidally acceptable bactericide. Such bactericides include, but are not limited to, validamycin, streptomycin, salmycin, envviromycin, 2-phenylphenol, thymol, 4-tert-amyphenol, 4-chloro-3-methylphenol, 4-chloro-2-

benzylphenol and 4-chloro-3,5-dimethylphenol, 1,2-benzisothiazol-3(2H)-one (BIT), carbendazim, chlorotoluron, mixtures of 5-chloro-2-methyl-4-isothiazolin-3-one with 2-methyl-4-isothiazolin-3-one, 2,2-dibromo-3-nitrilopropionamide (DBNPA), fluometuron, 3-iodo-2-propenylbutyl carbamate (IPBC), isoproturon, 2-n-octyl-4-isothiazolin-3-one (OIT), prometryn, and propiconazole.

According to one embodiment, the at least one pesticidal active ingredient is at least one biopesticide. Suitable biopesticides include those derived from natural products (naturally occurring), such as animals, plants, bacteria (e.g., Howler from AgBiome), and certain minerals. According to one embodiment, the at least one biopesticide is synthetic (not naturally occurring). The at least one biopesticide may be used to control pests or the microorganisms that control pests. Suitable biopesticides include, but are not limited to, the biopesticides registered as biopesticide active ingredients with the United States Environmental Protection Agency. Such suitable biopesticides include, but are not limited to, cold pressed neem oil, *Ulocladium oudemansii*, lavandulyl senecioate, calcium acetate, 2-methyl-1-butanol, *Trichoderma asperellum*, *Trichoderma gamsii*, laminarin food use fungicide, (E,Z)-7,9-Dodecadien-1-yl acetate, abscisic acid, (Z,Z,E)-7,11,13-hexadecatrienal, coat protein gene of plum pox virus, homobrassinolide, *Chenopodium ambrosioides*, *Trichoderma hamatum* isolate, (Z,Z,E)-3,8,11-tetradecatrienyl acetate, cry1Ac in MON 87701, *Bacillus thuringiensis Vip3Aa20*, hydrogenated catmint oil, n-tetradecyl acetate, iron HEDTA, heptyl butyrate, sodium ferric EDTA, oriental mustard seed, Z-7-tetradecened-2-one, L-Lactic Acid, *Pasteuria usgae*, *Candida oleophila* strain O, yeast, trimethylamine, indole, L-carvone, fox urine, calcium lactate, *Chenopodium ambrosioides*, b. *firmus*, *Bacillus thuringiensis Cry1A.105*, *Bacillus thuringiensis Cry2Ab2*, vipcot, vip3Aa19, E,E-9,11-tetradecadienyl acetate, *Bacillus thuringiensis modified Cray 3*, R-octanol, indole-3-acetic acid, sorbitol octanoate, methyl eugenol, potassium silicate, ammonium nonanoate, culeure, ammonium bicarbonate, black pepper oil, iperine, citronellol, glycerol monocaprylate, propylene glycol, lysophosphatidylethanolamines, corn gluten meal, dipotassium phosphate, sucrose octanoate esters, silver nitrate, formic acid, xanthine, verbenone, fish oil, kaolin, canola oil, potassium dihydrogen phosphate, maple lactone, anthraquinone, acetic acid, iron phosphate, polyoxin D zinc salt, dihydroazadirachtin, and mixtures thereof.

According to one embodiment, the at least one pesticidal active ingredient is a plant growth regulator. Suitable plant growth regulators include, but are not limited to, ancymidol, azoluron, chlorflorene-methyl, flurprimidol, forchlorfenuron, indolylbutyric acid, mefluidide, 1-naphthylacetamide, 1-naphthylacetic acid, 2-naphthylacetic acid (ester), paclobutrazol, thidiazuron, 3-CPA, 4-CPA, BAP, butifos, tribufos, butralin, chlorflorene, clofencet, cyclanilide, daminozide, dicamba, dikegulac sodium, dimethipin, chlorfenethol, etacelasil, ethephon, ethchlorzate, fenoprop, 2,4,5-TP, fluoridamid, flurprimidol, flutriafol, guazatin, imazalil, karetazan, kinetin, lactidichlor-ethyl, maleic hydrazide, naptalam, quinmerac, sintofen, tetcyclacis, triiodobenzoic acid, triapenthenol, triazethan, tribufos, trinexapac-ethyl, uniconazole, protham and gibberillic acid, gibberellins and combinations thereof.

As provided herein, an at least one additional component is water, citric acid, potassium hydroxide, cocodimethylamine, polyoxyethylene sorbitan monolaurate, at least one emulsifier compound, at least one antifoaming compound, at least one antimicrobial agent, at least one buffer, at least one

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corrosion inhibitor, at least one dispersing agent, at least one dye, at least one fragrant, at least one freezing point depressant, at least one neutralizing agent, at least one odorant, at least one penetration aid, at least one sequestering agent, at least one spray drift control agent, at least one spreading agent, at least one stabilizer, at least one sticking agent, at least one viscosity-modifying additive, at least one preservative, at least one plasticizer, at least one hydrophobic barrier, at least one UV protectant, at least one water soluble solvent, or any combination thereof.

According to one embodiment, the at least one additional component is one or more surfactant. Such a surfactant may be one or more of an ionic and a non-ionic surfactant. Such surfactants may be used as an emulsifier, dispersant, solubilizer, wetter, penetrant, protective colloid, or for other purposes. Suitable ionic surfactants for use with the pesticidal composition described herein may include anionic surfactants such as alkali, alkaline earth or ammonium salts of sulfonates, sulfates, phosphates, carboxylates, and mixtures thereof. Suitable sulfonates include, but are not limited to, alkylarylsulfonates, diphenylsulfonates, alpha-olefin sulfonates, lignin sulfonates, sulfonates of fatty acids and oils, sulfonates of ethoxylated alkylphenols, sulfonates of alkoxyated arylphenols, sulfonates of condensed naphthalenes, sulfonates of dodecyl- and tridecylbenzenes, sulfonates of naphthalenes and alkylnaphthalenes, sulfosuccinates or sulfosuccinamates. Suitable sulfates include, but are not limited to, sulfates of fatty acids and oils, of ethoxylated alkylphenols, of alcohols, of ethoxylated alcohols, or of fatty acid esters. Suitable phosphates include, but are not limited to, phosphate esters. Suitable carboxylates include, but are not limited to, alkyl carboxylates, and carboxylated alcohol or alkylphenol ethoxylates. Suitable anionic surfactants include, but are not limited to, sulfates and sulfonates.

According to one embodiment, the at least one additional component is one or more nonionic surfactant. Suitable nonionic surfactants include, but are not limited to, alkoxyates, N-substituted fatty acid amides, amine oxides, esters, sugar-based surfactants, polymeric surfactants, and mixtures thereof. Suitable alkoxyates include, but are not limited to, compounds such as alcohols, alkylphenols, amines, amides, arylphenols, fatty acids or fatty acid esters which have been alkoxyated with from 1 to 50 molar equivalents of an alkoxyating agent such as, for example, ethylene oxide (EO), propylene oxide (PO), or a combination thereof.

According to another embodiment, the one or more non-ionic surfactant is an alcohol initiated EO/PO block copolymer such as a butanol initiated EO/PO block copolymer, which may also be known as a polyalkylene glycol monobutyl ether, a poly(ethylene glycol-co-propylene glycol) monobutyl ether, or a propylene oxide ethylene oxide polymer monobutyl ether. According to one embodiment, the butanol initiated EO/PO block copolymer may have a degree of ethoxylation of from about 20 to about 30 and a degree of propoxylation of from about 20 to about 30. Suitable examples of EO/PO block copolymers include, but are not limited to, Toximul® 8320 available from Stepan (Northfield, Ill.), Termul® 5429 available from Huntsman International LLC (The Woodlands, Tex.), Tergitol™ XD available from Dow Chemical (Midland, Mich.), and Ethylan™ NS 500LQ available from AkzoNobel (Chicago, Ill.).

According to one embodiment, the surfactant is one or more quaternary ammonium surfactant. Suitable quaternary ammonium surfactants include ARQUAD 2C-75 (available from Akzo Nobel, Chicago, Ill.), quaternary ammonium chloride salts, cocodimethyl and dicocodimethylammonium

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chloride, and other coco-substituted quaternary ammonium salts. According to one embodiment, the surfactant is one or more alkyl, primary, secondary or tertiary amine. According to one embodiment, the surfactant is one or more amine with alkyl groups or a linear alkenyl groups with a carbon number of 8 to 20. According to one embodiment, the surfactant is derived from natural oil or fat. According to one embodiment, the surfactant, includes one or more an oxyalkylene groups. Suitable amines include, but are not limited to, any of the ARMEEN aliphatic amines (available from Akzo Nobel), mono-long-chain alkylamine, for example, bis(2-hydroxyethyl)cocoamine, bis(2-hydroxyethyl)-tallowamine, bis(2-hydroxyethyl)oleylamine, and bis(2-hydroxyethyl) laurylamine, polyoxyalkylenated long-chain alkylamine, for example, bis(polyoxyethylene(EOp=3 to 30))cocoamine, bis(polyoxyethylene(EOp=3 to 30))tallowamine, bis(polyoxyethylene(EOp 3 to 30))oleylamine, bis(polyoxyethylene(EOp=3 to 30))laurylamine, bis(polyoxyethylene(EOp=3 to 30))palmstearylamine, bis(polyoxyethylene(EOp=3 to 10))polyoxypropylene-(POp=3 to 10)cocoamine, and bis (polyoxyethylene (EOp=3 to 10)polyoxypropylene (POp=3 to 10))tallowamine. In the compounds described above, EOp represents an average addition mole number of ethylene oxide, and POp represents an average addition mole number of propylene oxide.

According to one embodiment, the surfactant includes one or more chelating agent and one or more cation compound. Combinations of oxalic acid and citric acid may be utilized due to the combination's action as an effective permeabilizer without exhibiting toxicity on exposure. According to one embodiment, citric acid lowers oxalic acid toxicity and also functions as a permeabilizing agent.

According to one embodiment, the at least one additional component is at least one inert formulation ingredient. Suitable inert formulation ingredients include, but are not limited to, dispersants, surfactants and wetting agents. Additional composition components may also include, for example, one or more ingredients, which may be dissolved or dispersed in the composition and may be selected from acaricides, algicides, antifeedants, avicides, bird repellents, chemosterilants, defoliants, desiccants, disinfectants, herbicide safeners, flow agents, insect attractants, insect repellents, mammal repellents, mating disrupters, molluscicides, nematocides, plant activators, plant growth regulators, rodenticides, semiochemicals, synergists, virucides, antifouling agents, antimicrobial agents, buffers, corrosion inhibitors, dispersing agents, dyes, fragrances, freezing point depressants, neutralizing agents, odorants, penetration aids, sequestering agents, spray drift control agents, spreading agents, stabilizers, sticking agents, viscosity-modifying additives, preservatives, plasticizers, hydrophobic barriers, UV protectants, water soluble solvents and mixtures thereof.

According to one embodiment, the at least one additional component is at least one flow agent. Suitable flow agents include, but are not limited to, clays such as kaolin, talc, diatomaceous earth and propylene glycol.

According to one embodiment, the at least one additional component is a component to aid in forming a film on the surface of a crop plant. Such a film forming component may include, but are not limited to, cellulose acetate, cellulose acetate-succinate, cellulose acetate phthalate, hydroxyethyl-cellulose, hydroxypropylcellulose, carboxymethylcellulose, carboxyethylcellulose, chitosan, methylcellulose, ethyl cellulose, propylcellulose, butylcellulose, alkylcelluloses, phthalate and acetate esters of cellulose, hypromellose, hypromellose acetate succinate, hypromellose phthalate, xanthan gum, guar gum, gellan gum, gum arabic, cara-

geenan, alginic acid (and its salts), acacia, tragacanth, polyvinyl acetate, polyvinyl alcohol, polyvinylpyrrolidone, polyvinylacetate phthalate, methacrylic-acrylic acid copolymer and its alkyl esters or ethers and combinations thereof.

According to one embodiment, the at least one additional component increases the adhesion of a particular component or composition to the target area (i.e., leaf). The adhesion component may aid in rain fastness. Suitable adhesion components include the film forming compounds provided herein, polyvinyl alcohol, polyvinylpyrrolidone, or any combination thereof.

According to one embodiment, the at least one additional component increases the penetration of the pesticidal active ingredient. Suitable penetration components include, but are not limited to, oranosilicone-based surfactants.

According to one embodiment, the at least one additional component is a component that aids in drift control. Suitable drift control components include, but are not limited to, viscoelastic polymers.

According to one embodiment, the at least one additional component is a suitable base or pH control agent. According to a particular embodiment, the base is potassium hydroxide.

According to one embodiment, the at least one additional component is at least one permeabilizing agent. According to one embodiment, the at least one permeabilizing agent is at least one chelating agent. According to one embodiment, the at least one chelating agent is ethylenediaminetetraacetic acid (EDTA), diethylenetriaminepentaacetic acid (DTPA), (hydroxyethyl)ethylenediaminetriacetic acid (HEDTA), nitrilotriacetic acid (NTA), oxalic acid, citric acid, sugars, amino acids, organic diacids, diamines, alpha ketoacids, alphahydroxyacids, aminodiacids, amino tri-acids, amino tetra-acids, t-dol amines, organic polyacids (and their sodium, potassium, and ammonium salts), salts of maleic acid, malonic acid, tartaric acid, glycine, lactic acid, malic acid, succinic acid, dextrose, tris(hydroxymethyl)aminomethane, lactose, mannitol, glutaric acid, malic acid, succinic acid, glycerol, humic acid, fulvic acid, sorbic acid, sorbose, ethylene diamine, 1,2 diaminocyclohexane, trimethylenediamine, tetramethylenediamine, 1,2 diaminopropane, diethylenetriamine, triethylenetetramine, triaminodimethylamine, N-hydroxyethylethylenediamine, sodium polyphosphate, potassium polyphosphate, ammonium polyphosphate, sodium hexametaphosphate and mixtures thereof.

According to one embodiment, the at least one additional component is water conditioning agent. Suitable water conditioning agents include, but are not limited to, pH buffering agents.

According to one embodiment, the at least one additional is at least one component or chemical disclosed by Harris in U.S. Pat. No. 9,113,625 and Miles in U.S. Pat. No. 7,476, 646, the contents of which are each herein incorporated by reference.

A tank mix pesticidal composition is also provided. The tank mix composition may be applied directly to a target plant or pest. The tank mix pesticidal composition is particularly suited for direct application to at least one crop plant, the surrounding soil, turf, seed, or a combination thereof. The tank mix pesticidal composition reduces or eliminates at least one or more fungus, bacteria, weed, insect or other pest.

According to one embodiment, the tank mix pesticidal composition includes at least one pesticidal active ingredient, at least one pesticidal active ingredient and at least one

organic solvent. According to one embodiment, the tank mix pesticidal composition further includes water, spray oil, or a combination thereof.

The at least one organic solvent included in the tank mix pesticidal composition is any organic solvent as provided herein. Suitable organic solvents include, but are not limited to, propylene carbonate, N-methylpyrrolidone, benzyl acetate, benzyl butyrate, benzyl propionate, diethyl malonate, 3-methoxy-3-methyl-1-butanol, dimethyl benzyl carbinol acetate, phenyl ethyl alcohol, terpinyl acetate, benzyl benzoate, methyl salicylate, hexyl acetate, benzyl butyrate, diethyl malonate, ethyl acetate, phenoxy ethyl iso-butyrate, terpineol, geranyl acetate, linalyl acetate, ethyl 2-methyl butyrate, propylene glycol diacetate, dipropylene glycol monomethyl ether, or any combination thereof. According to a particular embodiment, the at least one organic solvent included in the pesticidal composition is benzyl acetate. According to a particular embodiment, the at least one organic solvent included in the pesticidal composition is propylene carbonate. According to a particular embodiment, the at least one organic solvent included in the pesticidal composition is N-methylpyrrolidone. According to one embodiment, the at least one active ingredient in the tank mix pesticidal composition at least one pesticidal active ingredient as provided herein. According to one embodiment, the at least one organic solvent in the tank mix is benzyl acetate, benzyl alcohol, phenyl ethyl alcohol, hexyl acetate, terpinyl acetate, terpineol, geranyl acetate, linalyl acetate, propylene glycol diacetate, benzyl propionate, dipropylene glycol monomethyl ether, propylene carbonate, 3-Methoxy-3-methyl-1-butanol, N-methylpyrrolidone, or any combination thereof.

The tank mix pesticidal compositions may be formed by combining at least one organic solvent as provided herein, at least one pesticidal active ingredient, and optionally at least one of any variety of additional components as provided herein. According to a particular embodiment, the at least one organic solvent, pesticidal active ingredient, water, spray oil, and any optional additional components may be added to a tank to form a tank mix pesticidal composition. At least one de-foamer composition may be added to the pesticidal composition at the time of mixing.

According to a particular embodiment, a tank mix pesticidal composition is provided that includes from typically about 0.01% w/w to about 5% w/w of at least one pesticidal active ingredient. According to a particular embodiment, a tank mix pesticidal composition is provided that includes from typically about 0.2% w/w to about 2 w/w % of at least one organic solvent. According to a particular embodiment, a tank mix pesticidal composition is provided that includes from typically about 80% w/w to about 99% of a water, spray oil (e.g., phytobland or paraffinic oil), or a combination thereof.

The tank mix pesticidal compositions or as provided herein may be applied to plant leaves as foliar sprays, or to plant shoots, or to the surrounding soil or to seed. According to one embodiment, the tank mix pesticidal compositions as provided herein may be applied via a sprayer attached to a tank containing the pesticidal composition. According to one embodiment, the tank mix pesticidal compositions as provided herein may be applied via orchard sprayers, ground sprayers, row crop applicators, electrostatic sprayers, drone sprayers, helicopter sprayers, airplaner sprayers, or any other sprayer combatible for spraying the compositions as provided herein on agricultural and horticultural crops.

According to one embodiment, the tank mix pesticidal compositions as provided herein may be applied via an

irrigation system (chemigation). According to one embodiment, the tank mix pesticidal compositions as provided herein may be applied as a drench treatment. According to one embodiment, the tank mix pesticidal compositions as provided herein may be applied as a wick system.

An alternative adjuvant composition is also provided. The adjuvant compositions as provided herein may be utilized to improve the efficacy of at least one active ingredient such as a fungicide, bactericide, herbicide, insecticide, pesticide, or biopesticide. According to one embodiment, the at least one pesticidal active ingredient may be sufficient to reduce, halt, or eliminate growth of various target pests such as fungi, bacteria, insects, or weeds. According to one embodiment, the adjuvant compositions as provided herein may be utilized to improve the efficacy of at least one pesticidal active ingredient such that the pesticidal active ingredient is more efficacious against at least one resistant pest such as a fungi, bacteria, insect or weed compared to when the active ingredient is utilized alone.

According to one embodiment, the adjuvant compositions include at least one organic solvent as provided herein. According to a particular embodiment, the at least one organic solvent included in the adjuvant composition is benzyl acetate. According to a particular embodiment, the at least one organic solvent included in the pesticidal composition is propylene carbonate. According to a particular embodiment, the at least one organic solvent included in the pesticidal composition is N-methylpyrrolidone. According to one embodiment, the at least one organic solvent included in the adjuvant composition is benzyl acetate, benzyl alcohol, phenyl ethyl alcohol, hexyl acetate, terpinyl acetate, terpineol, geranyl acetate, linalyl acetate, propylene glycol diacetate, benzyl propionate, dipropylene glycol monomethyl ether, propylene carbonate, 3-Methoxy-3-methyl-1-butanol, N-methylpyrrolidone, or any combination thereof.

According to one embodiment, the adjuvant compositions as provided herein may further at least one additional component as provided herein. Particularly suitable additional components include, but are not limited to, water, citric acid, potassium hydroxide, cocodimethylamine, polyoxyethylene sorbitan monolaurate (commercially available as Tween™ 21), at least one emulsifier compound, at least one antifoaming compound, at least one antimicrobial agent, at least one buffer, at least one corrosion inhibitor, at least one dispersing agent, at least one dye, at least one fragrant, at least one freezing point depressant, at least one neutralizing agent, at least one odorant, at least one penetration aid, at least one sequestering agent, at least one spray drift control agent, at least one spreading agent, at least one stabilizer, at least one sticking agent, at least one viscosity-modifying additive, at least one preservative, at least one plasticizer, at least one hydrophobic barrier, at least one UV protectant, at least one water soluble solvent, or any combination thereof.

According to one embodiment, the adjuvant composition includes typically at least from about 10% by weight (w/w) to about 98% by weight (w/w) of at least one organic solvent as provided herein. According to one embodiment, the adjuvant composition includes typically at least from about 20% by weight (w/w) of at least one organic solvent as provided herein. According to one embodiment, the adjuvant composition includes typically at least from about 30% by weight (w/w) of at least one organic solvent as provided herein. According to one embodiment, the adjuvant composition includes typically at least from about 40% by weight (w/w) of at least one organic solvent as provided herein. According to one embodiment, the adjuvant composition

includes typically at least from about 50% by weight (w/w) of at least one organic solvent as provided herein. According to one embodiment, the adjuvant composition includes typically at least from about 60% by weight (w/w) of at least one organic solvent as provided herein. According to one embodiment, the adjuvant composition includes typically at least from about 70% by weight (w/w) of at least one organic solvent as provided herein. According to one embodiment, the adjuvant composition includes typically at least from about 75% by weight (w/w) of at least one organic solvent as provided herein. According to one embodiment, the adjuvant composition includes typically at least from about 80% by weight (w/w) of at least one organic solvent as provided herein.

According to one embodiment, the adjuvant composition includes typically at least from about 2% by weight (w/w) to about 90% by weight (w/w) of at least one additional component as provided herein. According to one embodiment, the adjuvant composition includes typically at least from about 5% by weight (w/w) of at least one additional component as provided herein. According to one embodiment, the adjuvant composition includes typically at least from about 10% by weight (w/w) of at least one additional component as provided herein. According to one embodiment, the adjuvant composition includes typically at least from about 15% by weight (w/w) of at least one additional component as provided herein. According to one embodiment, the adjuvant composition includes typically at least from about 20% by weight (w/w) of at least one additional component as provided herein. According to one embodiment, the adjuvant composition includes typically at least from about 25% by weight (w/w) of at least one additional component as provided herein. According to one embodiment, the adjuvant composition includes typically at least from about 35% by weight (w/w) of at least one additional component as provided herein. According to one embodiment, the adjuvant composition includes typically at least from about 45% by weight (w/w) of at least one additional component as provided herein. According to one embodiment, the adjuvant composition includes typically at least from about 55% by weight (w/w) of at least one additional component as provided herein. According to one embodiment, the adjuvant composition includes typically at least from about 65% by weight (w/w) of at least one additional component as provided herein. According to one embodiment, the adjuvant composition includes typically at least from about 75% by weight (w/w) of at least one additional component as provided herein. According to one embodiment, the adjuvant composition includes typically at least from about 85% by weight (w/w) of at least one additional component as provided herein.

According to a particular embodiment, the adjuvant composition typically includes from about 0.0% w/w to about 2.0% w/w water. According to a particular embodiment, the adjuvant composition typically includes about 0.4% w/w water. According to a particular embodiment, the adjuvant composition typically includes from about 0.0% w/w to about 2.0% w/w citric acid. According to a particular embodiment, the adjuvant composition typically includes about 0.2% w/w citric acid. According to a particular embodiment, the adjuvant composition typically includes from about 0.0% w/w to about 2.0% w/w potassium hydroxide. According to a particular embodiment, the adjuvant composition typically includes about 0.2% w/w potassium hydroxide. According to a particular embodiment, the adjuvant composition typically includes from about 0.0% w/w to about 2.0% w/w cocodimethylamine. According to a par-

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ticular embodiment, the adjuvant composition typically includes about 0.2% w/w cocodimethylamine. According to a particular embodiment, the adjuvant composition typically includes from about 20% w/w to about 99% w/w of at least one organic solvent. According to a particular embodiment, the at least one organic solvent is benzyl acetate, propylene carbonate, N-methylpyrrolidone, or any combination thereof. According to a particular embodiment, the adjuvant composition typically includes about 74% w/w benzyl acetate. According to a particular embodiment, the adjuvant composition typically includes from about 10% w/w to about 70% w/w of at least one surfactant. According to one embodiment, the at least one surfactant is polyoxyethylene sorbitan monolaurate. According to a particular embodiment, the adjuvant composition typically includes about 25% w/w polyoxyethylene sorbitan monolaurate.

According to a particular embodiment, the organic solvent is applied at an effective rate of typically at least about 0.1 fluid ounces/acre. According to a particular embodiment, the organic solvent is applied at an effective rate of typically at least about 1 fluid ounces/acre. According to a particular embodiment, the organic solvent is applied at an effective rate of typically at least about 2 fluid ounces/acre. According to a particular embodiment, the organic solvent is applied at an effective rate of typically at least about 3 fluid ounces/acre. According to a particular embodiment, the organic solvent is applied at an effective rate of typically at least about 4 fluid ounces/acre. According to a particular embodiment, the organic solvent is applied at an effective rate of typically at least about 5 fluid ounces/acre. According to a particular embodiment, the organic solvent is applied at an effective rate of typically at least about 6 fluid ounces/acre. According to a particular embodiment, the organic solvent is applied at an effective rate of typically at least about 7 fluid ounces/acre. According to a particular embodiment, the organic solvent is applied at an effective rate of typically at least about 8 fluid ounces/acre. According to a particular embodiment, the organic solvent is applied at an effective rate of typically at least about 9 fluid ounces/acre. According to a particular embodiment, the organic solvent is applied at an effective rate of typically at least about 10 fluid ounces/acre. According to a particular embodiment, the organic solvent is applied at an effective rate of typically at least about 11 fluid ounces/acre. According to a particular embodiment, the organic solvent is applied at an effective rate of typically at least about 12 fluid ounces/acre. According to a particular embodiment, the organic solvent is applied at an effective rate of typically at least about 13 fluid ounces/acre. According to a particular embodiment, the organic solvent is applied at an effective rate of typically at least about 14 fluid ounces/acre. According to a particular embodiment, the organic solvent is applied at an effective rate of typically less than about 15 fluid ounces/acre.

According to one aspect, a plant is provided that includes a pesticidal composition, adjuvant composition, or a tank mix pesticidal composition on a surface thereof.

According to one aspect, a kit is provided. The kit includes a container that includes an adjuvant composition or pesticidal composition as provided herein. The kit may optionally include instructions for use.

A method of producing an adjuvant composition for use in combination with at least one pesticidal active ingredient is provided. The method includes the step of introducing at least one organic solvent to a container. The at least one organic solvent includes propylene carbonate, N-methylpyrrolidone, benzyl acetate, benzyl butyrate, benzyl propionate, diethyl malonate, 3-methoxy-3-methyl-1-butanol, dimethyl

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benzyl carbinol acetate, phenyl ethyl alcohol, terpinyl acetate, benzyl benzoate, methyl salicylate, hexyl acetate, benzyl butyrate, diethyl malonate, ethyl acetate, phenoxy ethyl iso-butyrate, terpineol, geranyl acetate, linalyl acetate, ethyl 2-methyl butyrate, propylene glycol diacetate, dipropylene glycol monomethyl ether, or any combination thereof. According to one embodiment, the at least one organic is benzyl acetate, benzyl alcohol, phenyl ethyl alcohol, hexyl acetate, terpinyl acetate, terpineol, geranyl acetate, linalyl acetate, propylene glycol diacetate, benzyl propionate, dipropylene glycol monomethyl ether, propylene carbonate, 3-Methoxy-3-methyl-1-butanol, N-methylpyrrolidone, or any combination thereof. The method optionally includes the step of introducing at least one additional component to the container. The at least one additional component includes water, citric acid, potassium hydroxide, cocodimethylamine, polyoxyethylene sorbitan monolaurate, at least one emulsifier compound, at least one antifoaming compound, at least one antimicrobial agent, at least one buffer, at least one corrosion inhibitor, at least one dispersing agent, at least one dye, at least one fragrant, at least one freezing point depressant, at least one neutralizing agent, at least one odorant, at least one penetration aid, at least one sequestering agent, at least one spray drift control agent, at least one spreading agent, at least one stabilizer, at least one sticking agent, at least one viscosity-modifying additive, at least one preservative, at least one plasticizer, at least one hydrophobic barrier, at least one UV protectant, at least one water soluble solvent, or any combination thereof. The method optionally includes the step of attaching at least one set of instructions to the container.

The at least one organic solvent is introduced to the container in an amount such that entire contents of the container may be added directly to a spray tank containing at least one pesticidal active ingredient. According to a particular embodiment, at least one organic solvent is benzyl acetate. According to one embodiment, the adjuvant composition increases the efficacy of the at least one pesticidal active ingredient by at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more compared to the at least one pesticidal active ingredient alone. According to one embodiment, the at least one pesticidal active ingredient is a fungicide, bactericide, herbicide, insecticide, pesticide, plant growth regulator, or biopesticide.

According to one embodiment, the adjuvant composition is added or mixed with the at least one active ingredient and water or spray oil in a tank at the time of spraying. According to one embodiment, limited amounts of the adjuvant composition are sufficient to increase the efficacy of the active ingredient by reducing the potential for phytotoxicity and leaving a portion of the active ingredient in solid form for residual activity.

A method of preparing a pesticidal composition is also provided. The method includes the steps of combining, within a container, at least one organic solvent as provided herein, at least one pesticidal active ingredient as provided herein, and, optionally, or more additional components as provided herein.

A method of enhancing the efficacy of a pesticidal active ingredient is provided. The method includes the step of introducing at least one adjuvant composition to at least one pesticidal active ingredient. The at least one adjuvant composition includes at least one organic solvent selected from the group consisting of propylene carbonate, N-methylpyrrolidone, benzyl acetate, benzyl butyrate, benzyl propionate, diethyl malonate, 3-methoxy-3-methyl-1-butanol, dimethyl

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benzyl carbinol acetate, phenyl ethyl alcohol, terpinyl acetate, benzyl benzoate, methyl sailcylate, hexyl acetate, benzyl butyrate, diethyl malonate, ethyl acetate, phenoxy ethyl iso-butyrate, terpineol, geranyl acetate, linalyl acetate, ethyl 2-methyl butyrate, propylene glycol diacetate, dipropylene glycol monomethyl ether, or any combination thereof. According to one embodiment, the at least one organic is benzyl acetate, benzyl alcohol, phenyl ethyl alcohol, hexyl acetate, terpinyl acetate, terpineol, geranyl acetate, linalyl acetate, propylene glycol diacetate, benzyl propionate, dipropylene glycol monomethyl ether, propylene carbonate, 3-Methoxy-3-methyl-1-butanol, N-methylpyrrolidone, or any combination thereof. According to one embodiment, the efficacy of the active ingredient is increased by at least about 5%, 10%, 15%, 20% 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more compared to application of the pesticidal active ingredient without the adjuvant composition. According to one embodiment, the at least one active ingredient is a fungicide, bactericide, herbicide, insecticide, pesticide, or biopesticide. According to one embodiment, the organic solvent is benzyl acetate.

A method of improving the quality of at least one target plant is provided. The target plant may be an agronomic plant, horticultural plant or economically significant turf grass. The method includes the step of applying a pesticidal composition to the surface of the at least one plant, the surrounding soil, turf, seed, or a combination thereof. The pesticidal composition that is applied includes at least one pesticidal active ingredient and at least one organic solvent. The at least one organic solvent may be propylene carbonate, N-methylpyrrolidone, benzyl acetate, benzyl butyrate, benzyl propionate, diethyl malonate, 3-methoxy-3-methyl-1-butanol, dimethyl benzyl carbinol acetate, phenyl ethyl alcohol, terpinyl acetate, benzyl benzoate, methyl sailcylate, hexyl acetate, benzyl butyrate, diethyl malonate, ethyl acetate, phenoxy ethyl iso-butyrate, terpineol, geranyl acetate, linalyl acetate, ethyl 2-methyl butyrate, propylene glycol diacetate, dipropylene glycol monomethyl ether, or any combination thereof. According to one embodiment, the at least one organic is benzyl acetate, benzyl alcohol, phenyl ethyl alcohol, hexyl acetate, terpinyl acetate, terpineol, geranyl acetate, linalyl acetate, propylene glycol diacetate, benzyl propionate, dipropylene glycol monomethyl ether, propylene carbonate, 3-Methoxy-3-methyl-1-butanol, N-methylpyrrolidone, or any combination thereof. The method improves the quality of at least one crop by reducing or eliminating at least one or more fungus, bacteria, weed, insect or other pest thereby reducing such pressures and improving the quality of the crop. The at least one active ingredient utilized in the method may be a fungicide, bactericide, herbicide, insecticide, pesticide, or biopesticide. According to a particular embodiment, the organic solvent is benzyl acetate.

A method of increasing yield of a crop is provided. The method includes the step of applying a pesticidal composition to the surface of the at least one crop plant, the surrounding soil, turf, seed, or a combination thereof. The pesticidal composition that is applied includes at least one pesticidal active ingredient and at least one organic solvent. The at least one organic solvent may be propylene carbonate, N-methylpyrrolidone, benzyl acetate, benzyl butyrate, benzyl propionate, diethyl malonate, 3-methoxy-3-methyl-1-butanol, dimethyl benzyl carbinol acetate, phenyl ethyl alcohol, terpinyl acetate, benzyl benzoate, methyl sailcylate, hexyl acetate, benzyl butyrate, diethyl malonate, ethyl acetate, phenoxy ethyl iso-butyrate, terpineol, geranyl

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acetate, linalyl acetate, ethyl 2-methyl butyrate, propylene glycol diacetate, dipropylene glycol monomethyl ether, or any combination thereof. According to one embodiment, the at least one organic is benzyl acetate, benzyl alcohol, phenyl ethyl alcohol, hexyl acetate, terpinyl acetate, terpineol, geranyl acetate, linalyl acetate, propylene glycol diacetate, benzyl propionate, dipropylene glycol monomethyl ether, propylene carbonate, 3-Methoxy-3-methyl-1-butanol, N-methylpyrrolidone, or any combination thereof. The method reduces or eliminates at least one or more fungus, bacteria, weed, insect or other pest thereby increasing crop yield. The at least one active ingredient utilized in the method may be a fungicide, bactericide, herbicide, insecticide, pesticide, or biopesticide. According to a particular embodiment, the organic solvent is benzyl acetate.

A method of reducing post blossom fruit drop is provided. The method includes the step of applying a pesticidal composition to the surface of the at least one crop plant, the surrounding soil, turf, seed, or a combination thereof. The pesticidal composition that is applied includes at least one pesticidal active ingredient and at least one organic solvent. The at least one organic solvent may be propylene carbonate, N-methylpyrrolidone, benzyl acetate, benzyl butyrate, benzyl propionate, diethyl malonate, 3-methoxy-3-methyl-1-butanol, dimethyl benzyl carbinol acetate, phenyl ethyl alcohol, terpinyl acetate, benzyl benzoate, methyl sailcylate, hexyl acetate, benzyl butyrate, diethyl malonate, ethyl acetate, phenoxy ethyl iso-butyrate, terpineol, geranyl acetate, linalyl acetate, ethyl 2-methyl butyrate, propylene glycol diacetate, dipropylene glycol monomethyl ether, or any combination thereof. According to one embodiment, the at least one organic is benzyl acetate, benzyl alcohol, phenyl ethyl alcohol, hexyl acetate, terpinyl acetate, terpineol, geranyl acetate, linalyl acetate, propylene glycol diacetate, benzyl propionate, dipropylene glycol monomethyl ether, propylene carbonate, 3-Methoxy-3-methyl-1-butanol, N-methylpyrrolidone, or any combination thereof. The method reduces post blossom fruit drop by reducing or eliminating at least one or more fungus, bacteria, weed, insect or other pest. The at least one active ingredient utilized in the method may be a fungicide, bactericide, herbicide, insecticide, pesticide, or biopesticide. According to a particular embodiment, the organic solvent is benzyl acetate.

Although specific embodiments of the present invention are herein illustrated and described in detail, the invention is not limited thereto. The above detailed descriptions are provided as exemplary of the present invention and should not be construed as constituting any limitation of the invention. Modifications will be obvious to those skilled in the art, and all modifications that do not depart from the spirit of the invention are intended to be included with the scope of the appended claims.

Example 1

Fungicide Testing (FRAC Group 11)

Throughout the Examples provided herein, the designation "ADJ1" refers to benzyl acetate and the designation "ADJ2" refers to alkylene carbonate. Throughout the Examples, the organic solvent compositions (e.g., Benzyl Acetate Composition) include about 70% w/w of the stated organic solvent and about 30% of an agriculturally acceptable surfactant such as, for example, polyoxyethylene sorbitan monolaurate.

Petri dish tests (i.e., zone of inhibition tests) were conducted which demonstrated that formulations containing the

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adjuvant compositions ADJ1 or ADJ2 in addition to a FRAC Group 11 fungicide, azoxystrobin (Quadris®), have at least greater than 25% reduction in radial growth on a resistant strain of *Didymella* sp. (gummy stem rot) compared to treatment with azoxystrobin (Quadris®) alone. The results are presented in Table 1 below.

TABLE 1

Quadris ® rate (fluid oz/Acre)	Quadris ® rate	Additive	Adjuvant Rate (fluid oz/Acre)	24 hours, cm	48 hrs, cm	72 hrs, cm
11	.5x	ADJ1	4	15.9	42	47.1
8	.37x	ADJ1	4	16.6	43.3	49.5
5.5	.25x	ADJ1	4	18.1	49	59.9
2.6	.12x	ADJ1	4	17.9	58	70.9
11	.5x	ADJ2	4	14.6	49.1	57.8
8	.37x	ADJ2	4	20.3	45.4	53.5
5.5	.25x	ADJ2	4	21.9	48.5	61.3
2.6	.12x	ADJ2	4	22.4	46.5	59.6
22	1x	none	0	24.9	54.3	72.3
16.5	.75x	none	0	26.6	56.5	74.1
11	.5x	none	0	24.8	59.9	75.4
5.5	.25x	none	0	26	63.5	85.5
0	0x	Untreated/control	0	26.9	74.1	88

Example 2

Fungicide Testing (FRAC Group 11)

Petri dish tests (i.e., zone of inhibition tests) were conducted to compare the efficacy of azoxystrobin against both sensitive/susceptible and resistant strains of *Didymella* sp. (gummy stem rot). To treat the susceptible strain, the equivalent of 2 oz/acre of ADJ1 was shown to give comparable control to a FRAC Group 11 fungicide, azoxystrobin, alone (below effective rate for ADJ1). To treat the resistant strain, adding 2 oz of ADJ1 was shown to give twice as much growth reduction compared to azoxystrobin alone at equivalent rates. Control of resistant strain was shown to be comparable to azoxystrobin on susceptible strains. ADJ1 alone showed some efficacy but not at a commercial level control. A calcium complex did not add to growth reduction. The results are presented in Tables 2 and 3.

TABLE 2

Adjuvant Rate (fl. oz/acre)	Quadris ® Rate (fluid oz/acre)	Name	18 hours, cm	24 hours, cm	36 hours, cm	48 hours, cm
0	0	Untreated/Control - sensitive	23.9	37.2	52.6	64.2
8	0	Calcium	19.6	25.1	42.9	51.9
		Complex + ADJ1				
0	14.0	Quadris ® 1x	10.1	11.8	25.8	31.9
0	12.75	Quadris ® 0.75x	10.9	13.6	24.1	27.2
0	7.0	Quadris ® 0.5x	10.4	13.9	16.6	25
0	3.5	Quadris ® 0.25x	17.6	13.5	20.5	20.6
8	0	ADJ1 1x	15.7	21.3	40.6	50.1
6	0	ADJ1 0.75x	15.7	26.1	38.8	48.1
4	0	ADJ1 0.5x	17	23.7	32.3	46
2	0	ADJ1 0.25x	19.9	24.9	39.3	47.9
4	14	ADJ1 0.5x + Quadris ® 1x	8.4	13.9	19.1	24.6
4	12.75	ADJ1 0.5x + Quadris ® 0.75x	9	12.4	19.5	28.2
4	7.0	ADJ1 0.5x + Quadris ® 0.5x	14	16.1	25.5	31
4	3.5	ADJ1 0.5x + Quadris ® 0.25x	14.5	17.5	24.1	27.6

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TABLE 3

Adjuvant Rate (fl. oz/acre)	Quadris ® Rate (fluid oz/acre)	Name	18 hours, cm	24 hours, cm	36 hours, cm	48 hours, cm
0	0	Untreated - resistant	29.5	38.4	51.6	66.4
4	0	Calcium Complex + ADJ1	19.9	24.9	40.3	50.1
0	14.0	Quadris ® 1x	22.6	32	44.3	57.7
0	12.75	Quadris ® 0.75x	29.7	37.5	49.4	62.5
0	7.0	Quadris ® 0.5x	30.1	37.4	50.9	64.2
0	3.5	Quadris ® 0.25x	30	38.4	49.6	64.5
8	0	ADJ1 1x	16.7	10.9	35.1	44.4
6	0	ADJ1 0.75x	18.9	23.4	35.4	45.9
4	0	ADJ1 0.5x	22.6	28.5	44.3	53.9
4	0	ADJ1 0.25x	20.6	30.2	42	51.2
4	14	ADJ1 0.5x + Quadris ® 1x	13.5	9.7	21.9	24.9
4	12.75	ADJ1 0.5x + Quadris ® 0.75x	13	12	19.6	28.2
4	7.0	ADJ1 0.5x + Quadris ® 0.5x	18.5	17.6	28.4	35
4	3.5	ADJ1 0.5x + Q Quadris ® 0.25x	18.6	19.7	31.1	39.7

Example 3

Fungicide Testing (FRAC Group 11)

Petri dish tests (i.e., zone of inhibition tests) were conducted to compare four rates of three adjuvant compositions (ADJ1, ADJ2, and ADJ3 (N-methylpyrrolidone)) on sensitive and resistant strains of *Didymella* at a fixed treatment rate with a FRAC Group 11 fungicide, azoxystrobin. All three adjuvant compositions were shown to provide excellent control of resistant strain at the lowest azoxystrobin label rates. The results are presented in Tables 4 (sensitive strain) and 5 (resistant strain) below.

TABLE 4

azoxystrobin Rate (fl. oz/acre)	Additive	Additive Rate	12 hours, cm	24 hours, cm	36 hours, cm
Untreated	0	0	24.9	41.4	60.5
Sensitive					
22	0	0	15.8	22.9	31.9
16	0	0	13.4	20	28
11	0	0	12.5	16.9	23.6
5.5	0	0	15.3	24.9	34.3
7	ADJ1	8	11.3	16.8	21
7	ADJ1	4	15.6	22	28.9
7	ADJ1	2	12.8	16	21
7	ADJ1	1	13.8	19.5	25
0	ADJ1	0	24	41.1	54
7	ADJ2	8	15.1	21.8	27.5
7	ADJ2	4	14.6	21.5	30
7	ADJ2	2	15.5	24	31.1
7	ADJ2	1	17.4	25.3	32
0	ADJ2	0	24.3	39.6	53.6
7	ADJ3	8	15	21.1	27.6
7	ADJ3	4	14.9	21.8	29.6
7	ADJ3	2	17.3	24.9	30.1
7	ADJ3	1	17.1	25.1	32.6
0	ADJ3	0	22.8	41	54.1

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TABLE 5

azoxystrobin Rate (fl. oz/ acre)	Additive	Additive Rate	12 hours, cm	24 hours, cm	36 hours, cm
Untreated	0	0	31	49	70
Resistant					
22	0	0	22	28	35
16	0	0	29	47	68
11	0	0	30	48	70
5.5	0	0	31	48	70
7	ADJ1	8	15	22	29
7	ADJ1	4	18	25	35
7	ADJ1	2	17	23	35
7	ADJ1	1	19	27	41
0	ADJ1	0	25	30	56

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then treated with: (a) a FRAC Group 11 fungicide, azoxystrobin alone; (b) azoxystrobin plus ADJ1; (c) azoxystrobin plus ADJ2; (d) azoxystrobin plus a typical commercial adjuvant (commercially available as Ad-Spray™); and (e) azoxystrobin plus chlorothalonil and commercial adjuvant (commercially available as Ad-Spray™). Disease severity ratings were taken and illustrated improved disease control of resistant strain with ADJ1 and ADJ2 compared to treatment with azoxystrobin alone, azoxystrobin plus commercial adjuvant, and azoxystrobin plus chlorothalonil and commercial adjuvant, especially at lower rates of azoxystrobin. Severity rating were based on multiple counts of active fungal infection sites on the leaves of the plants. The results are summarized in Table 6 below.

TABLE 6

Treatment	azoxystrobin application rate	Adjuvant	Adjuvant Rate (fl. oz/acre)	One Week Rating	Two Week Rating	Third Week Rating	Fourth Week Rating
Untreated	0	None	none	11.8	18.4	42.6	64.9
azoxystrobin	22	None	6	1.1	0.2	4.8	16.3
azoxystrobin	11	None	6	13.3	16.7	42	65.7
azoxystrobin	5.5	None	6	9.2	16.9	63.3	82.8
azoxystrobin	22	ADJ1	6	0.5	0.6	5.1	8.4
azoxystrobin	11	ADJ1	6	0.6	0	0	3.2
azoxystrobin	5.5	ADJ1	6	4.4	1.5	10.9	18.7
azoxystrobin	22	Ad-Spray™	4	0.5	0.6	11	9.5
azoxystrobin	11	Ad-Spray™	4	6.4	0	10.2	23.6
azoxystrobin	5.5	Ad-Spray™	4	9.3	3.4	39	60.6
azoxystrobin	22	ADJ2	6	3.9	1.2	1.5	5.1
azoxystrobin	11	ADJ2	6	9.1	0.2	2.4	9
azoxystrobin	5.5	ADJ2	6	7.7	0.4	11.5	20.6
azoxystrobin	22	Ad-Spray™	4	0	0	48	64.6
plus chlorothal							
azoxystrobin	11	Ad-Spray™	4	1	0	4	42.9
plus chlorothal							
azoxystrobin	5.5	Ad-Spray™	4	8.7	0	10.1	58.8
plus chlorothal							

TABLE 5-continued

azoxystrobin Rate (fl. oz/ acre)	Additive	Additive Rate	12 hours, cm	24 hours, cm	36 hours, cm
7	ADJ2	8	19	22	29
7	ADJ2	4	19	26	39
7	ADJ2	2	21	31	45
7	ADJ2	1	20	29	41
0	ADJ2	0	26	38	58
7	ADJ3	8	12	27	35
7	ADJ3	4	16	27	40
7	ADJ3	2	17	27	45
7	ADJ3	1	19	24	41
0	ADJ3	0	27	39	60

Example 4

Fungicide Testing (FRAC Group 11)

Watermelon *Didamella* (Gummy Stem Rot) Field Test

Watermelon plants were inoculated with an azoxystrobin resistant strain of *Didymella*. The watermelon plants were

Example 5

Fungicide Testing (FRAC Group 11)

Watermelon *Didymella* (Gummy Stem Rot) Field Test

Watermelon plants were grown in normal field conditions subject to infection by natural strains of *Didymella*. The watermelons were then treated with: (a) a FRAC Group 11 fungicide, azoxystrobin alone; (b) ADJ1 alone; (c) azoxystrobin plus ADJ2; (d) azoxystrobin plus a typical commercial adjuvant (commercially available as Ad-Spray™); and (e) azoxystrobin plus chlorothalonil and commercial adjuvant. Disease severity ratings were taken and illustrated improved disease control of natural strains with ADJ1 compared to treatment with azoxystrobin alone or azoxystrobin plus commercial adjuvant, and equivalent control to azoxystrobin plus chlorothalonil and commercial adjuvant. Some disease reduction was observed with ADJ1 only. The results are summarized in Table 7 below.

TABLE 7

	azoxystrobin Application Rate (fl. oz/acre)	Adjuvant	Adjuvant Rate (fl. oz/acre)	One Week Rating	Two Week Rating	Third Week Rating	Fourth Week Rating
Untreated	0	none	none	0	14.4	96.1	100
azoxystrobin	11	none	6	1.1	0	49.5	76.8
azoxystrobin	5.5	none	6	2.1	6.6	59.4	80.8
azoxystrobin	11	Ad- Spray™	4	0	0	12.1	30.9
azoxystrobin	5.5	Ad- Spray™	4	0.9	1.5	54.4	76.4
Adjuvant Only	0	ADJ1	6	1.1	8	38.9	48.7
azoxystrobin	11	ADJ1	6	0.2	0.2	0.9	3.4
azoxystrobin	5.5	ADJ1	6	6.8	2.2	17.8	25.1
azoxystrobin plus chlorothalonil	11	Ad- Spray™	4	0	0	0.6	5.9

Example 6

Fungicide Testing (FRAC Group 11)

Soybean *Cercospora* (Frog Eye Leaf Spot) Field Test

Soybean plants were grown in normal field conditions subject to infection by natural strains of *Cercospora* (known to be resistant to azoxystrobin) then treated with: (a) a FRAC Group 11 fungicide, azoxystrobin alone; (b) azoxystrobin plus ADJ1; (c) azoxystrobin plus ADJ2; (d) azoxystrobin plus a typical commercial adjuvant (commercially available as Preference®). Disease severity ratings were taken and illustrated improved disease control with ADJ1 and ADJ2 compared to treatment with azoxystrobin alone or azoxystrobin plus commercial adjuvant. Duncan Multiple Range treatment of the data (indicated by “a,” “b,” “c,” etc.) confirmed statistically valid different between the treatment results. The results are summarized in Table 8 below.

TABLE 8

	Severity Ratings											
	August 31/Upper leaf		September 7/higher Leaf		September 7/bottom leaf		September 7/middle leaf		September 13/bottom leaf		September 20/higher leaf	
Untreated	13.5	a	21.5	a	69.3		31.6	a	29	a	17.3	a
Azoxystrobin 0.5x	10.3	ab	16.3	a	35.5	b	3.2	c	18	b	11.3	b
Azoxystrobin 0.5x + ADJ1	1	c	4	b	38.8	b	1.9	c	5.3	d	3.3	c
Azoxystrobin 0.5x + ADJ2	1.5	c	2.3	b	36	b	1.7	c	1.5	d	1	c
Azoxystrobin 0.5x + Commercial Adjuvant	8	b	6.5	b	40.3	b	3.4	c	12	c	9.3	b

Example 7

Fungicide Testing (FRAC Group 11)

Corn Maydis (Southern Corn Blight) and Southern Corn Rust Field Test

Corn plants were grown in normal field conditions subject to infection by natural strains known to be resistant to azoxystrobin then treated with: (a) FRAC Group 11 fungicides, azoxystrobin plus picoxystrobin alone; (b) azoxystrobin plus picoxystrobin plus ADJ1; (c) azoxystrobin

plus ADJ2; and (d) azoxystrobin plus picoxystrobin plus a typical commercial adjuvant (commercially available as Preference®). Disease severity ratings were taken and illustrated improved disease control with ADJ1 and ADJ2 compared to treatment with azoxystrobin alone or azoxystrobin plus commercial adjuvant. Duncan Multiple Range treatment of the data confirms statistically valid different between the treatment results (indicated by “a,” “b,” “c,” etc.) in Table 9 below.

TABLE 9

Days after Treatment	Application Rate (fl. oz/acre)	Severity Rating																					
		21	22	23	24	25	26	27	28	29	30	31											
(DAT)																							
Untreated	none	40.2	a	100	a	100	a	84.8	a	55.2	a	100	a	100	a	91.2	a	76.4	a	70.8	a	67.2	a
Azoxystrobin + Picoxystrobin 0.5x	7	0.2	b	93.7	a	70.6	b	79.4	a	17.6	b	69.8	b	80.9	b	66.4	b	66.6	a	50.7	ab	25.4	b
Azoxystrobin + Picoxystrobin 0.5x ADJ1	7	0.2	b	54.6	c	24	c	13	c	1.4	c	3.5	e	4.5	d	6.1	d	5.6	c	2.8	c	3.7	d
Azoxystrobin + Picoxystrobin 0.5x ADJ2	6																						
Azoxystrobin + Picoxystrobin 0.5x Commercial Adjuvant	7	0	b	62.4	bc	25.3	c	10.5	c	1.5	c	3.1	e	2.4	d	3.4	d	5.1	c	2.3	c	3.6	d
	6																						
	7	0.2	b	98.5	a	74.7	b	59	ab	19.6	b	55.3	c	47.2	c	45.7	c	37.4	b	39.9	ab	27.1	b
	6																						

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Example 8

Fungicide Testing (FRAC Group 11)

Greenhouse Phytotoxicity (Crop Injury) Test

To evaluate the potential for ADJ1 to increase the risk for crop injury, watermelon, radish, and tomato plants were grown in the greenhouse and treated with a rate of a FRAC Group 11 fungicide, azoxystrobin, equal to four times the maximum label rate at weekly intervals. The data in tables 10, 11 and 12 confirm that azoxystrobin at 4x rate plus ADJ1 presented significantly less crop injury than azoxystrobin at 4x plus typical commercial adjuvant (commercially available as Preference®). Duncan Multiple Range treatment of the data (indicated by “a,” “b,” “c,” etc.) confirmed statistically valid different between the treatment results.

TABLE 10

	Azoxystrobin Application Rate (fl. oz/acre)	Adjuvant Rate (fl. oz/acre)	Severity Rating	Severity Rating	Severity Rating
Untreated - Watermelon	none	none	0	d	0
azoxystrobin 4x + ADJ1	56	24	3	cd	2.8
azoxystrobin 4x + Commercial Adjuvant	56	24	15.8	b	23.8

TABLE 11

	Azoxystrobin Application Rate (fl. oz/acre)	Adjuvant Rate (fl. oz/acre)	Severity Rating	Severity Rating	Severity Rating
Untreated - Radish	none	none	0	d	0
azoxystrobin 4x + ADJ1	56	24	2.3	cd	9.5
azoxystrobin 4x + Commercial Adjuvant	56	24	20.8	a	32.8

TABLE 12

	Azoxystrobin Application Rate (fl. oz/acre)	Adjuvant Rate (fl. oz/acre)	Severity Rating	Severity Rating	Severity Rating
Untreated - Tomato	none	none	0	d	0
azoxystrobin 4x + ADJ1	56	24	1.5	cd	2.8
azoxystrobin 4x + Commercial Adjuvant	56	24	16.8	b	20
					10.5
					c

Example 9

Fungicide Testing (FRAC Group 11)
Citrus Post Bloom Fruit Drop

Efficacy of azoxystrobin, with or without ADJ1 or alternate commercial adjuvant (Activator 90) was tested against post bloom fruit drop causal agent, *Colletotrichum acutatum*, on Valencia oranges grown in a commercial orchard in Lake Wales, Florida. The FRAC Group 11 fungicide, azoxystrobin (Abound®) was utilized and was commercially available from Syngenta. Trees were spaced 12.5 feet apart on 25 feet wide beds and treated with foliar sprays with a single D6 nozzle on a handboom at a volume of 100 GPA weekly for four weeks. *Colletotrichum*-infected flowers were counted weekly. The trees treated with the higher rate of the combination of azoxystrobin and ADJ1 had the fewest infected flowers. Infected flower counts were significantly highest on the untreated orange trees but lowest for trees treated with the high rate of azoxystrobin with ADJ1. The results are summarized in Table 13.

TABLE 13

Days after treatment (DAT)	Number of Infected Flowers					
	0	7	15	20	27	36
Untreated	33	33.4	30.8	30.4	29.7	22.4
azoxystrobin (Abound ®) 15.5 fl. oz plus adjuvant (Activator 90)	26.6	27.5	11.1	8.3	5.9	4.5

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TABLE 13-continued

Days after treatment (DAT)	Number of Infected Flowers					
	0	7	15	20	27	36
azoxystrobin (Abound ®) 15.5 fl. oz plus ADJ1	29.6	29.1	7.5	3.1	2.1	1.5

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Example 10

Fungicide Testing (FRAC Group 3)
Wheat Stripe Rust Control and Yield

Wheat was grown in normal field conditions subject to infection by natural strains known to be resistant to tebuconazole (demethylation inhibitor, dicofenoconazole, a triazole, and member of Fungicide Resistance Action Committee (FRAC) Group 3). The wheat was treated with: (a) tebuconazole (commercially available as Folicur®) plus commercial adjuvant ((commercially available as Preference®); (b) tebuconazole plus ADJ1; (c) a pre-formulated adjuvant composition of tebuconazole plus ADJ1. Disease severity ratings were taken and illustrated improved disease control with ADJ1 as well as ADJ1 pre-formulated with tebuconazole compared to treatment with tebuconazole plus commercial adjuvant. Duncan Multiple Range (DMR) treatment of the data (indicated by "a," "b," "c," etc.) confirms statistically valid difference between the treatment results. To assess the effect on yield, the wheat was harvested and weighed. The results are summarized in the Tables 14 and 15 below.

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TABLE 14

		21 Days After Treatment			28 Days		35 Days	
		Appli- cation	DMR		After Treatment		After Treatment	
Treatment		Rate (fl. oz/rate)	Severity Rating	Statistical Group	Severity Rating	DMR	Severity Rating	DMR
1 Untreated	none	29.7	a		49	a	64.3	a
2 Tebuconazole non-ionic surfactant	2 4	10.2	b		20.3	b	33.2	b
3 Tebuconazole ADJ1	2 6	1.2	c		3.3	c	10.9	c
4 Tebuconazole plus ADJ1 (pre-formulated)	2	0.4	c		2.2	c	8.7	c

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TABLE 15

Treatment	Yield Pounds	DMR Statistical Group	Yield Bushels	DMR Statistical Group	Percent Improvement vs NIS
1 Untreated	6.4	c	25.1	c	
2 Tebuconazole plus non-ionic surfactant	12.55	b	49.2	b	
3 Tebuconazole plus ADJ1	14.78	ab	57.9	ab	18%
4 Tebuconazole plus ADJ1 (pre-formulated)	15.98	a	62.6	a	27%

Example 11

Fungicide Testing (FRAC Group 3)

Peanuts *Sclerotinia* Control

Peanuts were grown in the field using typical pesticidal practices and treated with Provost® Opti (commercially available) which is a mixture of FRAC Group 3 fungicides, tebuconazole and prothioconazole. The addition of ADJ1 statistically reduced the presence of *Sclerotinia*, generally referred to as White Mold which is a significant disease in peanuts compared to Provost combined with a commercially available adjuvant (commercially available as Induce®). Duncan Multiple Range (DMR) treatment of the data (indicated by “a,” “b,” “c,” etc.) confirms statistically valid difference between the treatment results. The data is presented in Table 16 below.

TABLE 16

Treatment No.	Treatment Name	Application Rate (fl. oz/acre)	Severity Rating	Statistical Group
1	Untreated	none	18.13	a
2	Provost ® Opti ADJ1	8	4.37	c
3	Provost ® Opti ADJ1	6	6.77	bc
4	Provost ® Opti ADJ1	4	9.72	ab
5	Provost ® Opti Induce ®	8	12.33	ab
		6		

Example 12

Peanut Late Leaf Spot Control

Fungicide Testing (FRAC Groups 7 and 11)

Peanuts were grown in the field using typical pesticidal practices and treated with Priaxor® which is a fungicidal mixture of pyraclostrobin and fluxapyroxad (corresponding to Fungicide Resistance Action Committee (FRAC) Groups 7 and 11). The addition of the organic solvent adjuvant, ADJ1, statistically reduced the presence of *Cercosporidium*, commonly called Late Leaf Spot, a significant fungal disease in peanuts. Duncan Multiple Range (DMR) treatment of the data (indicated by “a,” “b,” “c,” etc.) confirms statistically valid difference between the treatment results (at 90%) when the organic solvent adjuvant, ADJ1, is combined with the FRAC Groups 7 and 11 fungicides. The results are summarized in Table 17 below.

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TABLE 17

Treatment Number	Fungicide	Application Rate (fl. oz/acre)	Severity Rating	Statistical Group
1	Untreated	none	360.758	a
2	Priaxor ® ADJ1	6	241.830	d
3	Priaxor ® ADJ1	6	247.484	cd
4	Priaxor ® ADJ1	4	239.317	d
5	Priaxor ® Induce	6	268.748	c
		6		

Example 13

Fungicide Testing (FRAC Group 11)

Petri dish tests (i.e., zone of inhibition tests) were conducted to compare the efficacy of azoxystrobin (a strobilurin, quinone outside inhibitor (QOI), and member of Fungicide Resistance Action Committee (FRAC) Group 11) against resistant strains of *Didymella* sp. (gummy stem rot). Various organic solvents (adjuvants) were utilized at varying rates (6 fluid ounces/acre or 3 ounces/acre). The results are summarized in Table 18 and show the organic solvents provide a substantial resistant strain growth reduction compared to a FRAC Group 11 active ingredient utilized alone.

TABLE 18

Treatment Name	Radial growth (cm) at 48 hours	Quadris ® Rate (fluid oz per acre)	Adjuvant Rate (fluid oz per acre)	Improvement at same rate of Quadris ® (%)
Untreated	28.9	0	0	
Quadris ® only	18.9	12	0	
Quadris ® only	25.9	6	0	
40 Benzyl Acetate Composition plus Quadris ®	5.8	12	6	70%
Benzyl Acetate Composition plus Quadris ®	8.0	6	6	69%
Benzyl Acetate Composition only	20.9	0	6	28%
45 Benzyl Alcohol Composition plus Quadris ®	4.8	12	6	75%
Benzyl Alcohol Composition plus Quadris ®	6.6	6	6	74%
Benzyl Alcohol Composition only	25.1	0	6	13%
50 Phenyl Ethyl Alcohol Composition plus Quadris ®	4.7	12	6	75%
Phenyl Ethyl Alcohol Composition plus Quadris ®	5.8	6	6	77%
Phenyl Ethyl Alcohol Composition only	20.1	0	6	30%
55 Hexyl Acetate Composition plus Quadris ®	6.4	12	6	66%
Hexyl Acetate Composition plus Quadris ®	7.9	6	6	69%
Hexyl Acetate Composition only	26.7	0	6	8%
60 Terpinyl Acetate Composition plus Quadris ®	4.8	12	6	75%
Terpinyl Acetate Composition plus Quadris ®	6.6	6	6	74%
Terpinyl Acetate Composition only	18.9	0	6	35%
65 Terpineol Composition plus Quadris ®	4.5	12	6	76%
Terpineol Composition plus Quadris ®	5.9	6	6	77%

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TABLE 18-continued

Treatment Name	Radial growth (cm) at 48 hours	Quadris ® Rate (fluid oz per acre)	Adjuvant Rate (fluid oz per acre)	Improvement at same rate of Quadris ® (%)
Terpineol Composition only	20.8	0	6	28%
Geranyl Acetate	7.6	12	6	60%
Composition plus Quadris ®				
Geranyl Acetate	9.2	6	6	64%
Composition plus Quadris ®				
Geranyl Acetate	26.9	0	6	7%
Composition only				
Linalyl Acetate Composition	4.3	12	6	77%
plus Quadris ®				
Linalyl Acetate Composition	10.8	6	6	58%
plus Quadris ®				
Linalyl Acetate Composition	18.9	0	6	35%
only				
Propylene Glycol Diacetate	4.3	12	6	78%
Composition plus Quadris ®				
Propylene Glycol Diacetate	9.8	6	6	62%
Composition plus Quadris ®				
Propylene Glycol Diacetate	19.9	0	6	31%
Composition only				
0.5x Benzyl alcohol	4.2	12	3	78%
Composition plus Quadris ®				
0.5x Benzyl alcohol	6.8	6	3	74%
Composition plus Quadris ®				
0.5x Benzyl Acetate	8.3	12	3	56%
Composition plus Quadris ®				
0.5x Benzyl Acetate	9.8	6	3	62%
Composition plus Quadris ®				

Example 14

Fungicide Testing (FRAC Group 3)

Petri dish tests (i.e., zone of inhibition tests) were conducted to compare the efficacy of tebuconazole (demethylation inhibitor, dicofenoconazole, a triazole, and member of Fungicide Resistance Action Committee (FRAC) Group 3) against resistant strains of *Rhizoctonia solani* (brown patch, stem rot, and/or root rot). Various organic solvents (adjuvants) were utilized at 6 fluid ounces/acre and compared with just the active ingredient alone. The results are summarized in Table 19 and show the organic solvents provide a substantial resistant strain growth reduction compared to a FRAC Group 3 active ingredient utilized alone.

TABLE 19

Treatment Name	Radial growth (cm) at 48 hours	% Reduction in Radial Growth	Improvement at same rate of Tebuconazole (%)
Untreated	15.72	n/a	n/a
Tebuconazole only (4 oz/acre)	7.8	50%	n/a
Tebuconazole only (2 oz/acre)	10.72	32%	n/a
Benzyl Propionate Composition plus Tebuconazole (4 oz/acre)	3.58	77%	53%
Benzyl Propionate Composition plus Tebuconazole (2 oz/acre)	4.02	74%	134%
Benzyl Propionate Composition only	5.13	67%	n/a
DPM Glycol Ether Composition plus Tebuconazole (4 oz/acre)	3.25	79%	57%
DPM Glycol Ether Composition plus Tebuconazole (2 oz/acre)	4.17	73%	131%

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TABLE 19-continued

Treatment Name	Radial growth (cm) at 48 hours	% Reduction in Radial Growth	Improvement at same rate of Tebuconazole (%)
DPM Glycol Ether Composition only	15.75	0%	n/a
Benzyl Alcohol Composition plus Tebuconazole (4 oz/acre)	3.47	78%	55%
Benzyl Alcohol Composition plus Tebuconazole (2 oz/acre)	4.5	71%	124%
Benzyl Alcohol Composition only	5.85	63%	n/a
Propylene Carbonate Composition plus Tebuconazole (4 oz/acre)	3.38	78%	56%
Propylene Carbonate Composition plus Tebuconazole (2 oz/acre)	4.57	71%	123%
Propylene Carbonate Composition only	14.72	6%	n/a
Benzyl Acetate Composition plus Tebuconazole (4 oz/acre)	3.63	77%	53%
Benzyl Acetate Composition plus Tebuconazole (2 oz/acre)	5.52	65%	104%
Benzyl Acetate Composition only	10.68	32%	35%
3-Methoxy-3-methyl-1-butanol	6.08	61%	22%
Composition plus Tebuconazole			
3-Methoxy-3-methyl-1-butanol	8.83	44%	38%
Composition plus Tebuconazole			
3-Methoxy-3-methyl-1-butanol	14.78	6%	28%
Composition only			
N-methyl pyrrolidone Composition plus Tebuconazole	4.82	69%	38%
N-methyl pyrrolidone Composition plus Tebuconazole	10.38	34%	7%
N-methyl pyrrolidone Composition only	15.92	-1%	n/a

Example 15

Herbicide Burndown Testing

Burndown (BD) testing was conducted on Morning Glory (MG), Goosegrass (GG), Florida Beggarweed (FB) with an herbicide, Gramoxone® (paraquat—N,N'-dimethyl-4,4'-bipyridinium dichloride). Various amounts of Crop Oil Concentrate™ (COC) and an organic solvent adjuvant, Benzyl Acetate (BA), were utilized in conjunction with the herbicide. Crop Oil Concentrate™ (COC) is commonly used to enhance the deposition and spreading of herbicides on the leaf surface. The Crop Oil Concentrate™ (COC) included paraffinic petroleum oil, tall oil fatty acids and alkylphenol ethoxylates. The results are summarized in Table 20 and show a substantial increase in burndown rate when the organic solvent adjuvant is combined with an herbicide. The results also show a substantial increase in burndown rate compared to the combination of herbicide and Crop Oil Concentrate™ (COC).

TABLE 20

Treatment	BD (MG) (%)	BD (GG) (%)	BD (FG) (%)	Improvement at same rate of paraquat, BD (MG) (%)	Improvement at same rate of paraquat, BD (GG) (%)	Improvement at same rate of paraquat, BD (FG) (%)
Paraquat only (2 pts/acre)	81	100	100			
Paraquat (2 pts/acre) plus Benzyl Acetate Composition (6 oz/acre)	88	100	100	9%	0%	0%
Paraquat (2pts/acre) plus Benzyl Acetate Composition (12 oz/acre)	94	100	100	16%	0%	0%
Paraquat (2 pts/acre) plus Benzyl Acetate Composition (24 oz/acre)	91	100	100	12%	0%	0%
Paraquat (2 pts/acre) plus COC (25 oz/acre)	55	71	75	-32%	-29%	-25%
Paraquat (1 pt/acre)	51	72	64			
Paraquat (1 pt/acre) plus Benzyl Acetate Composition (6 oz/acre)	78	84	78	77%	83%	77%
Paraquat (1 pt/acre) plus Benzyl Acetate Composition (12 oz/acre)	81	85	80	80%	84%	79%
Paraquat (1 pt/acre) plus Benzyl Acetate Composition (24 oz/acre)	75	82	83	74%	81%	82%
Paraquat (1 pt/acre) plus COC (25 oz/acre)	44	65	60	43%	64%	59%
Paraquat (0.5 pt/acre)	50	60	47			
Paraquat (0.5 pt/acre) plus Benzyl Acetate Composition (6 oz/acre)	84	88	85	68%	47%	81%
Paraquat (0.5 pt/acre) plus Benzyl Acetate Composition (12 oz/acre)	80	82	78	60%	37%	66%
Paraquat (0.5 pt/acre) plus Benzyl Acetate Composition (24 oz/acre)	88	87	90	76%	45%	91%
Paraquat (0.5 pt/acre) plus COC (25 oz/acre)	40	35	51	-20%	-42%	9%

Example 16

Fungicide Testing (FRAC Group 1)

Petri dish tests (i.e., zone of inhibition tests) were conducted to compare the efficacy of Topsin® (thiophanate methyl which is a thiophanate, methyl benzimidazole carbamate, and member of Fungicide Resistance Action Committee (FRAC) Group 1) against susceptible strains of *Botrytis cinera* (grey mold). The organic solvent adjuvant utilized was benzyl acetate. The non-ionic, low foaming surfactant/penetrant, LI 700®, was utilized in combination with the Topsin® in two of the treatments and applied at a rate of 32 fl oz/100 gallons. The effectiveness of benzyl acetate at an application rate of 6 oz/acre was compared with the effectiveness of LI 700® when combined with Topsin®. The results are summarized in Table 21 and show the organic solvent provides a substantial strain growth reduction compared to a FRAC Group 1 active ingredient utilized in combination with LI 700®.

TABLE 21

Treatment Name	Topsin ® Application Rate (oz/acre)	Radial growth (cm) at 48 hours	% Improvement in Disease Control vs. LI 700 ® at same rate of Topsin ®
Topsin ® and LI 700 ®	1.0	36	n/a
Topsin ® and LI 700 ®	0.5	39	n/a

TABLE 21-continued

Treatment Name	Topsin ® Application Rate (oz/acre)	Radial growth (cm) at 48 hours	% Improvement in Disease Control vs. LI 700 ® at same rate of Topsin ®
Benzyl Acetate plus Topsin®	1.0	9	75%
Benzyl Acetate plus Topsin ®	0.5	13	67%

Example 17

Fungicide Testing (FRAC Group 9)

Petri dish tests (i.e., zone of inhibition tests) were conducted to compare the efficacy of Scala® (pyrimethanil, an anilinopyrimidine and member of Fungicide Resistance Action Committee (FRAC) Group 9) against susceptible strains of *Botrytis cinera* (grey mold). The organic solvent adjuvant utilized was benzyl acetate. The non-ionic, low foaming surfactant/penetrant, LI 700®, was utilized in combination with the Topsin® in two of the trials and applied at a rate of 32 fl oz/100 gallons. The results are summarized in Table 22 and show the organic solvent, benzyl acetate (6 oz/acre), provides a substantial strain growth reduction compared to a FRAC Group 9 active ingredient utilized alone.

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TABLE 22

Treatment Name	Scala ® Application Rate (oz/acre)	Radial growth (cm) at 48 hours	% Improvement in Disease Control vs. LI 700 ® at same rate of Scala ®
Scala ® and LI 700 ®	18	17	n/a
Scala ® and LI 700 ®	9	23	n/a
Benzyl Acetate plus Scala ®	18	3	82%
Benzyl Acetate plus Scala ®	9	3	87%

Example 18

Fungicide Testing (FRAC Group 7)

Petri dish tests (i.e., zone of inhibition tests) were conducted to compare the efficacy of Fontelis® (penthiopyrad which is a carboxamide and member of Fungicide Resistance Action Committee (FRAC) Group 7) against susceptible strains of *Botrytis cinera* (grey mold). The organic solvent adjuvant utilized was benzyl acetate. The non-ionic, low foaming surfactant/penetrant, LI 700®, was utilized in combination with the Topsin® in two of the trials and applied at a rate of 32 fl oz/100 gallons. The results are summarized in Table 23 and show the organic solvent, benzyl acetate (6 oz/acre), provides a substantial strain growth reduction compared to a FRAC Group 7 active ingredient utilized alone.

TABLE 23

Treatment Name	Fontelis ® Application Rate (oz/acre)	Radial growth (cm) at 48 hours	% Improvement in Disease Control vs. LI 700 ® at same rate of Fontelis ®
Fontelis ® and LI 700 ®	24	25	n/a
Fontelis ® and LI 700 ®	12	30	n/a
Benzyl Acetate plus Fontelis ®	24	7	65%
Benzyl Acetate plus Fontelis ®	12	13	55%

Example 19

Fungicide Testing (FRAC Group M)

Petri dish tests (i.e., zone of inhibition tests) were conducted to compare the efficacy of the combination of Captan (phthalimide and member of Fungicide Resistance Action Committee (FRAC) Group M4) and Thiram (tetramethylthiuram disulfide, dithiocarbamate and member of Fungicide Resistance Action Committee (FRAC) Group M3) against susceptible strains of *Botrytis cinera* (grey mold). The organic solvent adjuvants utilized were benzyl acetate and DPM Glycol Ether. The non-ionic, low foaming surfactant/penetrant, LI 700®, was utilized in combination with the Captan/Thiram in two of the trials and applied at a rate of 32 fl oz/100 gallons. The results are summarized in Table 24 and show the organic solvents provide a substantial strain growth reduction compared to a FRAC Group M active ingredient utilized alone.

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TABLE 24

Treatment Name	Captan/Thiram Application Rate (lbs/acre)	Radial growth (cm) at 48 hours	% Improvement in Disease Control vs. LI 700 ® at same rate of Captan/Thiram
Captan/Thiram and LI 700 ®	6.0/4.4	6	n/a
Captan/Thiram and LI 700 ®	3.0/2.2	12	n/a
Benzyl Acetate (6.0 oz/acre) plus Captan/Thiram	6.0/4.4	3	50%
Benzyl Acetate (6.0 oz/acre) plus Captan/Thiram	3.0/2.2	11	8%

Example 20

Bactericide Testing

Petri dish tests (i.e., zone of inhibition tests) were conducted to compare the efficacy of a copper hydroxide bactericide (Kocide® 3000) against susceptible strains of *xanthomonas*. The organic solvent adjuvant utilized was benzyl acetate (BA) at a rate of 6 fluid oz/acre. A non-ionic surfactant (NIS) was utilized in combination with three of the trials at a rate of 32 fluid ounces per 100 gallons. The results are summarized in Table 25 and show the organic solvent provided a substantial bacterial strain growth reduction when compared with the non-ionic surfactant combined with the same rate of copper hydroxide bactericide.

TABLE 25

Treatment Name	Radial growth (cm) at 72 hours	Statistical Group	% Improvement in radial growth versus NIS with same rate of bactericide
Untreated	481.4	ab	
Non-Ionic Surfactant (NIS)	521.2	a	
Benzyl Acetate (BA)	458.5	ab	
Bactericide (4 lbs/acre) plus NIS	201.8	ghi	
Bactericide (2 lbs/acre) plus NIS	212.4	fgh	
Bactericide (1 lbs/acre) plus NIS	349.1	cd	
Bactericide (0.5 lbs/acre) plus NIS	407.7	bc	
Bactericide (4 lbs/acre) plus BA	140.4	ij	30%
Bactericide (2 lbs/acre) plus BA	181.8	g-j	14%
Bactericide (1 lbs/acre) plus BA	246.5	efg	29%
Bactericide (0.5 lbs/acre) plus BA	238	efg	42%

Example 21

Fungicide Testing (FRAC Group M)

Petri dish tests (i.e., zone of inhibition tests) were conducted to compare the efficacy of the combination of Captan (phthalimide and member of Fungicide Resistance Action Committee (FRAC) Group M4) and Thiram (tetramethylthiuram disulfide, dithiocarbamate and member of Fungicide Resistance Action Committee (FRAC) Group M3) against susceptible strains of *Botrytis cinera* (grey mold). Various organic solvent adjuvants were utilized at a rate of 6 oz/acre. The results are summarized in Table 26 and show the organic solvents provide a substantial strain growth reduction compared to the FRAC Group M active ingredient utilized alone.

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TABLE 26

Treatment Name	Captan/ Thiram Application Rate (lbs/acre)	Radial growth (cm) at 96 hours	% Improvement in Disease Control vs. Captan/ Thiram alone
Captan/Thiram	6.0/4.4	59	n/a
Benzyl Propionate plus Captan/Thiram	6.0/4.4	21	64%
Benzyl Propionate plus Captan/Thiram	4.5/3.3	36	39%
Benzyl Propionate plus Captan/Thiram	3.0/2.2	36	39%
Benzyl Butyrate plus Captan/Thiram	6.0/4.4	19	68%
Benzyl Butyrate plus Captan/Thiram	4.5/3.3	14	76%
Benzyl Butyrate plus Captan/Thiram	3.0/2.2	21	64%
Diethyl Malonate plus Captan/ Thiram	6.0/4.4	14	76
Diethyl Malonate plus Captan/ Thiram	4.5/3.3	11	81
Diethyl Malonate plus Captan/ Thiram	3.0/2.2	69	-17
3-Methoxy-3-methyl-1-butanol plus Captan/Thiram	6.0/4.4	16	73%
3-Methoxy-3-methyl-1-butanol plus Captan/Thiram	4.5/3.3	24	59%
3-Methoxy-3-methyl-1-butanol plus Captan/Thiram	3.0/2.2	50	15%
Benzyl Acetate plus Captan/Thiram	6.0/4.4	14	76%
Benzyl Acetate plus Captan/Thiram	4.5/3.3	15	75%
Benzyl Acetate plus Captan/Thiram	3.0/2.2	18	69%

Prophetic Example 22

Psyllid (Insect) Control

One or more of the organic solvents as provided herein would be combined with abamectin to treat citrus mites at a normal use rate according to the label. The one or more organic solvents would be effectively applied at a rate of about 6 fluid ounces/acre. A reduced population count would be observed at 3, 6, 9 days after treatment (DAT).

Prophetic Example 23

One or more of the organic solvents as provided herein would be combined with spinosad and sprayed on leaf miners at a normal use rate according to the label. The one

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or more organic solvents would be effectively applied at a rate of about 6 fluid ounces/acre. Reduced population could be expected to be observed at 3, 6, and 9 days after treatment (DAT).

The invention claimed is:

1. An adjuvant composition for increasing the efficacy of a herbicidal active ingredient, the adjuvant composition comprising:

benzyl acetate; and

N,N'-dimethyl-4,4'-bipyridinium dichloride,

wherein the benzyl acetate increases the efficacy of the N,N'-dimethyl-4,4'-bipyridinium dichloride compared to the N,N'-dimethyl-4,4'-bipyridinium dichloride alone.

2. An adjuvant composition for increasing the efficacy of a bactericide active ingredient, the adjuvant composition comprising:

diethyl malonate;

phthalimide; and

dithiocarbamate

wherein the diethyl malonate increases the efficacy of the phthalimide and dithiocarbamate compared to the phthalimide and dithiocarbamate alone.

3. A tank mix pesticidal composition comprising:

the adjuvant composition of claim 1; and

water, spray oil, or a combination of water and spray oil.

4. The tank mix pesticidal composition of claim 3, comprising:

from about 0.01% w/w to about 5% w/w of N,N'-dimethyl-4,4'-bipyridinium dichloride;

from about 0.2% w/w to about 2 w/w % benzyl acetate; and

from about 80% w/w to about 99% of the water, spray oil, or a combination of water and spray oil.

5. A tank mix pesticidal composition comprising:

the adjuvant composition of claim 2; and

water, spray oil, or a combination of water and spray oil.

6. The tank mix pesticidal composition of claim 5, comprising:

from about 0.01% w/w to about 5% w/w phthalimide;

from about 0.01% w/w to about 5% w/w dithiocarbamate;

from about 0.2% w/w to about 2 w/w % diethyl malonate; and

from about 80% w/w to about 99% of the water, spray oil, or a combination of water and spray oil.

* * * * *